

Poverty-Reducing Income Taxation with Endogenous Income^{*}

Oriol Carbonell-Nicolau[†]

February 2026

Abstract

This paper develops nonlinear income tax systems designed to reduce poverty within a framework where income is endogenously determined. Our approach to poverty reduction is applicable across a wide range of poverty measures and remains robust under varying economic conditions, including heterogeneous wage distributions, non-labor endowments, and diverse individual preferences. When negative marginal tax rates are deemed infeasible—potentially due to concerns about vertical equity—any poverty-reducing tax schedule must take the form of a fixed lump-sum subsidy for individuals whose after-tax incomes fall below the poverty line. Even when negative marginal tax rates are permissible, effective poverty-reducing tax policies must still rely on subsidization for the poor and non-positive marginal tax rates as essential features.

Keywords: poverty measurement, nonlinear income taxation, poverty-reducing taxation.

JEL classifications: D63, D71, I32.

1. Introduction

Unlike the traditional utilitarian approach prevalent in neoclassical economics, this paper employs a non-welfarist framework for policy evaluation. We explore the design of poverty-focused, nonlinear income tax systems, evaluating their effectiveness across diverse economic contexts.

The framework we develop emphasizes three core considerations of the underlying economic problem. First, we treat income as endogenous, recognizing that individuals adjust their earnings in response to tax policies, available non-labor income, and labor market opportunities. Second, we demonstrate that the impact of tax policy on poverty remains qualitatively consistent across varying economic conditions and heterogeneous individual preferences—elements that are either dynamic or unobservable. Third, our theoretical framework applies to a broad spectrum of established poverty measures, enhancing its practical relevance.

While the traditional welfarist approach requires the government to aggregate individual utility functions—unobservable theoretical constructs—into a social welfare function that

^{*}Constructive and insightful comments from the editors and the referee are gratefully acknowledged.

[†]Department of Economics, Rutgers University, 75 Hamilton St., New Brunswick, NJ 08901. E-mail: carbonell-nicolau@rutgers.edu.

makes explicit judgments about interpersonal utility comparisons (see [Boadway, 2012](#)), our poverty-focused framework provides a more concrete and measurable policy objective. This shift toward an observable and widely understood policy target offers theoretical clarity and practical advantages for policy design and evaluation.

Our analysis builds on the classical Mirrlees model of endogenous income ([Mirrlees, 1971](#)), extending it to incorporate two additional dimensions of heterogeneity beyond the standard variation in individual wage rates.

First, the framework allows for heterogeneity in individual preferences. Admissible preferences are represented by idiosyncratic utility functions defined over labor-consumption bundles, which are assumed to satisfy standard regularity conditions. A central assumption is the normality of consumption, which implies that consumption increases in response to lump-sum subsidies, holding other factors constant.

Second, the model introduces heterogeneity in non-labor income through individual-specific endowments. This variation captures differences in unearned income across individuals, further enriching the economic environment.

Poverty measurement has generated extensive scholarly literature encompassing diverse methodologies for complete and incomplete poverty orderings of income distributions. Despite this methodological variety, economists broadly agree that poverty measures must conform to certain fundamental principles.¹ Our theoretical framework builds upon four essential axioms from this literature: *Focus*, *Symmetry*, *Continuity*, and *Monotonicity*.

The Focus axiom stipulates that poverty measurement should respond exclusively to income changes among the poor population, remaining invariant to changes in non-poor incomes. Symmetry ensures that poverty measurement remains independent of individual characteristics beyond income, treating all individuals with equal income identically. The Continuity axiom requires that marginal changes in income levels produce correspondingly small changes in poverty measurements. Finally, the Monotonicity axiom establishes that *ceteris paribus*, any decrease in a poor person's income must result in an increased poverty measure.

These four foundational axioms, first formally articulated in [Sen's \(1976\)](#) seminal work, form the minimal requirements for our theoretical analysis. More precisely, we demonstrate that our results hold for *any* poverty measure adhering to these principles.

We consider a multidimensional tax policy space where an *income tax schedule* is a piecewise linear mapping from gross incomes to tax liabilities. These policies depend solely on observed gross income and cannot condition on unobserved individual characteristics such as wage rates, non-labor endowments, or preferences. Additionally, a valid income tax schedule must satisfy three fundamental properties—continuity, rank preservation, and monotonicity—while ensuring that tax liabilities never exceed gross incomes. Each property serves an important economic purpose in tax system design.

Continuity of the tax schedule ensures small income changes yield correspondingly small changes in tax liability, preventing cliff effects that distort behavior and violate horizontal equity.

Rank preservation ensures the relative income ordering of individuals remains unchanged post-taxation, maintaining the perceived fairness of the system.

Monotonicity embodies vertical equity: higher-income individuals face weakly higher tax liabilities. This precludes negative marginal tax rates—a constraint extensively studied

¹For a comprehensive review of these principles, see [Zheng \(2000\)](#).

in optimal taxation. While our main characterization of poverty-reducing schedules assumes monotonicity, an extended analysis relaxes it, revealing that negative marginal rates can emerge, consistent with related studies discussed later.

A *poverty-reducing tax schedule* ensures that, across all economic environments—regardless of non-labor endowment or wage distributions and individual preference profiles—the after-tax income distribution exhibits no more poverty than the tax-free distribution. This robustness makes poverty reduction a consistent policy goal despite dynamic or unobservable economic factors.

We provide a comprehensive characterization of poverty-reducing tax schedules: For a given poverty line $z > 0$, which serves as the threshold below which individuals are classified as “poor,” a tax schedule is poverty-reducing if and only if it provides a fixed lump-sum subsidy to individuals whose after-tax incomes fall below z .

This characterization rests on the following insight about how tax schedules interact with poverty levels. Consider any tax schedule that either fails to provide subsidies to the poor or implements a positive marginal tax rate. Such a schedule can increase poverty relative to a tax-free environment. This holds in a homogeneous economy, where identical individuals achieve higher consumption levels in the absence of taxation, thereby experiencing lower levels of poverty.

Beyond offering a clear and implementable criterion for tax policy design, this characterization yields several insights.

First, the poverty-reduction property defined in this paper is context-free: it cannot be contingent on resource constraints, which fluctuate with the underlying model fundamentals. Our analysis identifies the essential characteristics of poverty-reducing tax policies, providing a robust theoretical foundation upon which more specific, context-dependent policies can be built.

Second, the universality of our results extends beyond context-independence with respect to endowment and wage distributions to encompass essentially the entire class of admissible preference profiles. This feature stems from the pivotal role of homogeneous societies in our proofs: any tax schedule violating the uniform-subsidy characterization can be shown to increase poverty in a homogeneous environment using particular standard preferences. Our complete characterization holds as long as these standard functional forms are feasible, imposing no substantive restrictions on admissible taste heterogeneity.

This preference robustness is absent from inequality- and polarization-reducing tax design, which requires not only the classical single-crossing condition (Mirrlees, 1971) but additional order-preserving restrictions on the wage elasticity of income (Carbonell-Nicolau and Llavador, 2018, 2021a,b, 2025; Carbonell-Nicolau, 2019, 2024, 2025). Our framework is free of such constraints.

Third, we can establish a complete ordering of all poverty-reducing tax schedules based on a single criterion: the size of their lump-sum subsidies. Tax schedules that provide larger subsidies consistently yield lower poverty measurements, regardless of the specific poverty measure used, as long as it adheres to the four fundamental axioms of Focus, Symmetry, Continuity, and Monotonicity. This complete ordering enables straightforward comparisons between different poverty-reducing tax schedules beyond their poverty dominance relative to a tax-free environment.

Fourth, when the monotonicity requirement for tax schedules is relaxed, a broader class of poverty-reducing policies emerges: any tax schedule that subsidizes the poor and

maintains a non-increasing marginal tax rate for all after-tax incomes below the poverty line qualifies as poverty-reducing.

Removing the monotonicity constraint opens the door to subsidizing the poor through negative marginal tax rates. Under such a system, subsidies can increase as individuals earn more income, creating an incentive structure that rewards additional work. This approach aligns with findings in the broader literature, which we explore in greater detail below.

However, despite this expanded policy space, one fundamental principle from our earlier analysis remains unchanged: attempting to subsidize the poor while simultaneously imposing positive marginal tax rates is ineffective at reducing poverty.

Finally, we maintain rank preservation (requiring that higher gross income never yields lower net income) as a modeling assumption for normative reasons, but this condition is not essential to our characterization. In fact, downward-sloping segments of a tax schedule—where earning more gross income reduces net income—are economically irrelevant, since they are never chosen by rational agents, who can simply reduce labor supply to reach a better point on the budget constraint.

Beyond these core insights, we test the robustness of our characterization in three directions. First, we examine alternative poverty measurement frameworks to assess how our results depend on the adopted axiomatic structure. Second, we investigate institutional and informational constraints—such as endogenous poverty lines or multiple contested thresholds—that complicate real-world implementation. Third, we connect our outcome-based approach to normative principles from taxation theory, revealing both convergences and tensions.

Regarding poverty measurement, we analyze distribution-sensitive criteria that incorporate inequality concerns among the poor—building on [Shorrocks's \(1983\)](#) relative and absolute generalized Lorenz criteria, applied to censored income distributions. Here, the requirement of uniform lump-sum subsidies remains necessary: any non-fixed subsidy can increase poverty even under these refined measures, as demonstrated by homogeneous economies where inequality among the poor is absent. However, sufficiency fails—uniform subsidies alone no longer guarantee poverty reduction. When agents respond heterogeneously to a fixed subsidy, all poor incomes may rise in absolute terms while inequality among the poor simultaneously worsens. Under distribution-sensitive criteria, such trade-offs render post-tax distributions incomparable to the tax-free baseline, breaking the universal poverty reduction guarantee implied by our core axioms.

On informational and institutional constraints, we first address the case of multiple poverty lines. We show that requiring robustness across multiple poverty lines within a set $Z \subseteq \mathbb{R}_{++}$ imposes no additional structural restrictions beyond those in our main theorem. It merely requires that the subsidy domain be defined relative to $\sup Z$, the highest threshold under consideration. Building on this result, we can extend the analysis to endogenously determined poverty lines—such as thresholds defined as a fixed fraction of median income—with one important caveat. If poverty lines depend on post-tax incomes, a fundamental circularity arises whereby poverty status depends on the very tax policy being evaluated. However, this circularity dissolves when the threshold anchors to predetermined reference statistics (e.g., lagged medians). In such cases, our characterization for shifting thresholds implies that universal poverty reduction requires lump-sum subsidies for post-tax income levels below the maximum poverty line. This prescription preserves our core robustness desideratum—even when the poverty line calibrates to historical or projected fundamentals,

the actual economic environment at implementation (wage distributions, endowments, preferences) remains uncertain.

The critical role of endogenous income emerges most starkly when comparing our framework to its exogenous counterpart. With fixed incomes, poverty reduction requires only two mechanical conditions: (i) non-poor status preservation—taxation must not push non-poor individuals into poverty—and (ii) weak subsidization—every poor individual must receive a non-negative transfer. When income is endogenous, however, behavioral responses fundamentally reshape the policy space: only uniform lump-sum subsidies guarantee universal poverty reduction. Non-uniform subsidies induce substitution effects that may reduce post-tax income below its tax-free level for some agents. Thus, endogenizing labor supply transforms poverty alleviation from a mechanical redistribution problem into one constrained by incentive compatibility—a distinction with direct policy implications.

Finally, we bridge our framework to two alternative normative principles.

Adapting Young’s (1987; 1988; 1990) classical *equal sacrifice* principle from taxation theory, we identify conditions under which poverty-reducing subsidies deliver *equal benefit* (identical utility gains) to all individuals remaining poor post-subsidy. Under quasilinear preferences, uniform lump-sum subsidies implement equal benefit by eliminating income effects on labor supply. Conversely, equal benefit generally requires the absence of income effects for agents remaining poor—which highlights the contrast between outcome-based poverty reduction and utility-based fairness.

We also introduce *poverty preservation*—the requirement that taxation not reverse pre-existing poverty orderings across societies—as a cross-society analogue of rank preservation. While any poverty-preserving schedule must be poverty-reducing, the converse fails: uniform lump-sum subsidies satisfy universal poverty reduction yet may reverse poverty rankings across societies due to heterogeneous income effects on labor supply.

1.1. Literature on poverty and taxation

The literature on income taxation and poverty has diverged into two primary research trajectories. The first integrates poverty concerns into the classical utilitarian framework of optimal taxation, part of a broader scholarly effort to incorporate fairness principles into traditional social welfare theory (Fleurbaey and Maniquet, 2018). The second, which we follow, departs from traditional welfarism by placing poverty alleviation at the center of the analysis as an explicit policy objective.

Within the first tradition, Saez and Stantcheva (2016) develop a refined approach to optimal taxation using generalized social marginal welfare weights that can be concentrated among individuals below a poverty threshold. Rather than maximizing an explicit social welfare function, their framework evaluates local tax reforms, assigning higher weights to gains accruing to the poor while maintaining Pareto efficiency. Even with heavy poverty weighting, they find that optimal marginal tax rates below the poverty line remain positive—consistent with standard optimal tax results.²

However, recent work by Sher (2024) challenges the normative coherence of this methodology. Sher demonstrates that when marginal welfare weights depend on policy outcomes—such as an individual’s status relative to a poverty threshold—they generally fail to represent the gradient of any global social welfare function. Consequently, while

²Negative marginal rates can emerge in welfarist models under specific conditions such as fixed participation costs (Saez, 2002) or present bias (Lockwood, 2020).

such weights facilitate local reform comparisons, the implied social preferences may be intransitive, failing to aggregate into a coherent global ordering unless they satisfy “structural utilitarianism”—a condition requiring that weights depend solely on individuals’ utility levels rather than their positions in the income distribution.

The welfarist analyses in [Henry de Frahan and Maniquet \(2021\)](#) and [Maniquet and Neumann \(2021\)](#) avoid Sher’s critique by focusing on well-defined social preference orderings rather than income-contingent local weights.

[Henry de Frahan and Maniquet \(2021\)](#) bridge axiomatic fairness and optimal taxation by constructing a social welfare function based on poverty reduction and the principle of responsibility for preferences—the idea that agents with the same wage should be treated identically regardless of their labor choices. To resolve the potential conflict between poverty alleviation and Pareto efficiency, the authors propose a novel definition of poverty where an agent is considered poor only if their entire indifference curve lies below the poverty line. Their derived optimal tax formula features zero and negative marginal tax rates for low-income earners and exhibits significant “bunching” at the earnings level of the most hardworking minimum-wage households.

Maniquet et al. [Maniquet and Neumann \(2021\)](#) develop an antipoverty requirement that respects individual well-being by never requiring Pareto-inferior transfers. This requirement mandates that transfers from those above the poverty line to those below it are a social improvement only when both agents work the same hours. This axiom, together with Pareto and Independence, characterizes a social preference relation that is egalitarian according to a specific well-being index—the labor time a minimum-wage worker would need, along their own indifference curve, to reach the poverty line. Under the assumption that some minimum-wage worker’s after-tax income matches the poverty line, this index reduces to the pre-tax earnings level divided by the minimum wage—an observable measure. The authors show that any optimal tax scheme must feature negative marginal tax rates on very low incomes, providing a normative justification for in-work benefits.

Our framework avoids the Sher critique by treating poverty reduction as a primitive objective. We abandon welfarist aggregation entirely: poverty status is not mapped onto marginal social welfare weights; it is evaluated directly on the post-tax income distribution.

Our income-based approach excludes welfare costs of work effort—a normative choice that offers several analytical benefits.

First, it targets what taxpayers care about—visible income deprivation—which is an appropriate basis for claims on public resources ([Besley and Coate, 1995](#)).

Second, our model incorporates heterogeneous non-labor income, a dimension often omitted in welfarist axiomatic frameworks such as [Henry de Frahan and Maniquet \(2021\)](#) and [Maniquet and Neumann \(2021\)](#). Those models rely on a single well-being index derived from labor-leisure trade-offs, which complicates the inclusion of heterogeneous endowments because wage rates, preference parameters, and endowments cannot be separately identified from labor supply data alone. By focusing on the final income distribution, our approach fully integrates heterogeneous endowments without any identification challenges.

Third, our characterization requires no distributional knowledge—it holds universally across all wage, endowment, and preference distributions. In contrast, the optimal policy design in [Henry de Frahan and Maniquet \(2021\)](#) is sensitive to the underlying density of types and behavioral elasticities within a population. Unlike optimal tax approaches,

the evaluation criterion in [Maniquet and Neumann \(2021\)](#) does not require distributional data—though it relies on an identifying assumption about who earns the poverty-line income and abstracts from non-labor endowments.

Fourth, we deliver a complete classification of poverty-reducing tax schedules—identifying them as uniform lump-sum subsidies—and provide a complete ordering among them based on subsidy size. In contrast, optimal tax schedules in welfarist frameworks are often complex, non-linear functions that depend on the specific shape of social marginal welfare weights and the local distribution of types. Our approach reduces the policy space to a single parameter: the size of the uniform subsidy. Under any poverty measure satisfying the core axioms, a larger subsidy unambiguously reduces poverty.

Despite fundamental normative differences, our framework converges with [Henry de Frahan and Maniquet \(2021\)](#) and [Maniquet and Neumann \(2021\)](#) on a common policy implication: marginal tax rates on low incomes should be zero or negative.

We now review the literature that, like our paper, takes poverty alleviation as its explicit and primary policy objective.

[Besley and Coate \(1992\)](#) and [Besley and Coate \(1995\)](#) provide a positive foundation for income-based poverty analysis, arguing that poverty programs funded by general taxation reflect taxpayers' concern with visible income deprivation rather than recipients' unobserved leisure choices. This justifies evaluating anti-poverty policy directly on the income distribution—a premise our paper shares.

Besley and Coate further characterize cost-minimizing income maintenance programs designed to bring all individuals above a poverty threshold, proposing work requirements for low-wage workers as a means of reducing public expenditure.

[Bourguignon and Fields \(1990\)](#) examine the features of a poverty-minimizing anti-poverty budget using complete orderings of income distributions. They find that the optimal allocation is sensitive to the specific poverty index adopted and the initial income distribution. By contrast, our characterization does not optimize subject to a fixed resource constraint; instead, it asks which tax schedules reduce poverty universally, across all possible environments and poverty measures satisfying our four core poverty axioms. The answer—uniform lump-sum subsidies—is therefore invariant to the choice of poverty measure and the initial distribution of fundamentals.

[Kanbur et al. \(1994\)](#) provide a partial characterization of tax schedules aimed at minimizing an income-based poverty index of the additively separable form considered by [Atkinson \(1987\)](#). They find that optimal marginal tax rates should be negative at the lower end of the income distribution—aligning with our results—though their characterization does not impose universality across fundamentals.

Building on this work, [Pirttilä and Tuomala \(2004\)](#) extend the analysis to encompass mixed tax systems, which involve the simultaneous taxation of income and commodities, while maintaining the explicit social objective of poverty minimization. Their findings demonstrate that this approach justifies differentiated tax rates across commodities, challenging the conventional wisdom established by [Atkinson and Stiglitz \(1976\)](#), which advocates for uniform commodity taxation.

[Chambers and Moreno-Ternero \(2017\)](#) develop an axiomatic approach to poverty-sensitive taxation by building on Young's equal sacrifice principle ([Young, 1987, 1988, 1990](#)). They selectively remove certain axioms from Young's original framework—specifically, those axioms that prevent tax systems from exempting low-income individuals—enabling

a formal characterization of generalized equal-sacrifice tax rules that integrate poverty alleviation goals.

Finally, Carbonell-Nicolau (2026) also adopts a universal poverty reduction criterion, but focuses on distribution-sensitive poverty measures that need not satisfy the Monotonicity axiom. That paper completely characterizes poverty-reducing tax schedules within an exogenous income framework. The present paper shifts the focus to a non-nested class of poverty measures and extends the analysis to endogenous income, where behavioral responses fundamentally reshape the set of poverty-reducing policies.

The paper is structured as follows. Section 2 defines the class of poverty measures for which our results are valid. Section 3 introduces the endogenous income model and formulates the tax policy space. The main findings are presented in Section 4, where we discuss the implications of our results and develop heuristic insights into the logic behind the characterization of poverty-reducing tax schedules. Finally, Section 5 summarizes the main contributions and suggests directions for future research.

2. Poverty criterion

Our poverty criterion is censored first-order stochastic dominance, which encompasses all conventional poverty measures.

An *income distribution* is formally defined as an n -dimensional vector

$$\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}_+^n,$$

where each x_i denotes the income of individual i , and $n \in \mathbb{N}$ is the population size.

Let $(x_{[1]}, \dots, x_{[n]})$ be a non-decreasing rearrangement of the coordinates in $\mathbf{x}$:

$$x_{[1]} \leq \dots \leq x_{[n]}.$$

Given an income distribution $\mathbf{x} = (x_1, \dots, x_n)$ and a *poverty line* $z > 0$, we define the *poor* as the set of individuals i whose income x_i is less than or equal to z .

The distribution $\mathbf{x} = (x_1, \dots, x_n)$ *censored at* z , denoted as

$$\mathbf{x}^z = (x_1^z, \dots, x_n^z),$$

is defined by

$$x_i^z = \begin{cases} x_i & \text{if } x_i \leq z, \\ z & \text{if } x_i > z, \end{cases}$$

for each $i \in \{1, \dots, n\}$.

First-order stochastic dominance is a binary relation $\succsim_D$ over the set of income distributions. For any two distributions $\mathbf{x}$ and $\mathbf{y}$, we say that $\mathbf{x}$ dominates $\mathbf{y}$ (written as $\mathbf{x} \succsim_D \mathbf{y}$) if and only if

$$x_{[i]} \geq y_{[i]}, \quad \text{for each } i \in \{1, \dots, n\},$$

where $x_{[i]}$ and $y_{[i]}$ represent the i -th ordered elements of the respective distributions.

For a fixed poverty line $z > 0$, we establish a ranking system for income distributions using a poverty pre-order $\succsim_z$. This binary relation over the set of distributions is defined by

$$x \succsim_z y \Leftrightarrow y^z \succsim_D x^z,$$

where x^z and y^z represent the censored distributions of x and y , respectively, at the poverty line z .

The relation “ $x \succsim_z y$ ” should be interpreted as “the distribution x exhibits no less poverty than the distribution y ,” relative to the poverty line z .

The poverty pre-order $\succsim_z$ has been characterized in the poverty literature through a set of fundamental principles. Specifically, for fixed-size distributions, $\succsim_z$ is uniquely defined as the pre-order that simultaneously satisfies the following principles:

1. *Focus*: Changes in non-poor incomes do not affect poverty measurement.
2. *Symmetry*: Poverty measurement remains unchanged when individuals exchange incomes.
3. *Continuity*: Poverty measurement varies continuously with income levels.
4. *Monotonicity*: A decrease in a poor person’s income increases poverty.

A fundamental result regarding the characterization of $\succsim_z$ states that for any two income distributions x and y , we have $x \succsim_z y$ if and only if x exhibits no less poverty than y under all poverty measures that satisfy Focus, Symmetry, Continuity, and Monotonicity. This result was first established by [Atkinson \(1987\)](#) for the subclass of additively separable poverty measures. For a comprehensive discussion of these axioms and their implications, see [Zheng \(2000\)](#).

3. Endogenous income framework

Our model extends the foundational [Mirrlees \(1971\)](#) income taxation framework by incorporating both labor and non-labor income sources, as well as heterogeneous preferences.

Individual i ’s preferences over labor-consumption pairs $(l, x) \in [0, 1] \times \mathbb{R}_+$ are represented by a utility function $u_i(l, x)$, which is non-increasing in labor, l , and non-decreasing in consumption, x , and strictly decreasing in l and strictly increasing in x over $(0, 1) \times \mathbb{R}_{++}$. The quantity of labor hours, l , is normalized to the range between zero and one, where $l = 0$ represents no work and $l = 1$ represents full-time work.

The utility function u is continuous over its domain $[0, 1] \times \mathbb{R}_+$ and differentiable and strictly quasiconcave on the interior of this domain.³

³Two mild additional assumptions will be invoked in [Section 4.2.11](#): (i) leisure $1 - l$ is not inferior; and (ii) for every $x > 0$,

$$\lim_{l \rightarrow 0^+} (-\partial u_i(l, x) / \partial l) < \infty \quad \text{and} \quad \lim_{l \rightarrow 1^-} - \left(\frac{\partial u_i(l, x)}{\partial l} \right) / \left(\frac{\partial u_i(l, x)}{\partial x} \right) = +\infty. \quad (1)$$

The first condition in (1) rules out infinite marginal disutility at the onset of labor supply—an economically pathological case where agents would require infinite wages to work even an infinitesimal amount. The second condition in (1) states that the utility cost of an extra unit of work diverges to infinity as leisure time vanishes.

Furthermore, consumption, x , is assumed to be a normal good (i.e., it increases with income).

The set of all such utility functions is denoted by $\mathcal{U}$.

Each individual i is uniquely defined by their non-labor endowment $\omega_i \geq 0$ and wage rate $a_i > 0$, which together determine gross income $\omega_i + a_i l_i$, where l_i denotes hours worked.

An *income tax schedule* is defined as a continuous, rank-preserving, and non-decreasing function $t : \mathbb{R}_+ \rightarrow \mathbb{R}$, where $t(y)$ represents the tax liability associated with income level y , subject to the constraint that $t(y) \leq y$ for every $y \geq 0$, which ensures that tax liabilities never exceed pre-tax income levels, maintaining economic feasibility.

The monotonicity requirement embodies a weak form of vertical equity: individuals with higher income cannot face lower tax burdens. By design, this requirement explicitly precludes negative marginal tax rates, a concept that has garnered significant attention in the literature on nonlinear income taxation with endogenous income, both in welfarist and non-welfarist frameworks (see, e.g., Saez, 2002; Saez and Stantcheva, 2016; Boadway et al., 2002; Lockwood, 2020; Kanbur et al., 1994; Fleurbaey and Maniquet, 2006, 2007; Maniquet and Neumann, 2021; Henry de Frahan and Maniquet, 2021). Nevertheless, we demonstrate that this monotonicity constraint can be relaxed without compromising our core results. A detailed analysis of the implications of removing the monotonicity assumption is presented in [Appendix B](#) (see also the discussion in [Section 4.2.4](#)).

Rank preservation means that post-tax income $y - t(y)$ must be non-decreasing in pre-tax income y , preventing income rankings from being reversed through taxation.

Negative tax liabilities represent subsidies.

In this paper, we concentrate on *piecewise linear* tax schedules. Specifically, these schedules partition the domain $\mathbb{R}_+$ into a finite number of intervals, with the tax function assuming a linear form within each interval.

The set of all piecewise linear tax schedules is denoted by $\mathcal{T}$.

Remark 1 (Scope of admissible tax schedules). We restrict attention to income-contingent tax schedules—where an individual’s liability depends only on their own income—rather than vector-valued mappings that condition on the entire distribution. This reflects fundamental constraints on implementable tax policy under asymmetric information.

Standard optimal taxation theory addresses the government’s inability to observe ability directly by using income as a screening device to induce self-selection. Vector-valued taxes would undermine this mechanism by creating incentives to misreport income—not only to reduce one’s own tax burden but also to manipulate others’ liabilities. This interdependence transforms the tractable individual screening problem into a strategic game, compromising incentive compatibility while introducing equilibrium multiplicity and coordination failures.

Time-consistency challenges also intensify. Under a standard schedule, the government commits to a one-dimensional mapping from individual income to tax liability. Vector-valued taxation requires commitment to an n -dimensional policy across all possible equilibrium distributions, exponentially expanding the contingency set and multiplying opportunities for ex-post exploitation. Once individual choices are sunk, the government faces heightened incentives to renege, responding not just to individual characteristics but to the realized aggregate distribution—itsself a product of strategic responses to the policy.

Administratively, vector-valued systems require simultaneous processing of all returns to determine the aggregate distribution before computing individual liabilities, precluding the sequential processing that makes real-world tax administration feasible.

Finally, a foundational principle of tax fairness requires that observationally identical individuals face identical treatment. Vector-valued taxation violates this horizontal equity principle by imposing different liabilities on equivalent taxpayers based solely on others' incomes.

These considerations establish that income-contingent taxation, not vector-valued mappings, constitutes the relevant implementable policy space for normative tax analysis under asymmetric information.

An individual i with wage rate $a_i > 0$, non-labor income $\omega_i \geq 0$, and facing an income tax schedule t , chooses labor $l_i \in [0, 1]$ to solve the utility maximization problem

$$\max_{l_i \in [0,1]} u_i(l_i, \omega_i + a_i l_i - t(\omega_i + a_i l_i)),$$

where $\omega_i + a_i l_i - t(\omega_i + a_i l_i)$ is net income.

This problem has at least one solution due to continuity of u_i and compactness of the feasible set.

A *solution function* $l_i^{u_i}(\omega_i, a_i, t)$ solves this problem given ω_i , a_i , and t .

From $l_i^{u_i}$, define gross and net income functions:

$$y_i^{u_i}(\omega_i, a_i, t) = \omega_i + a_i l_i^{u_i}(\omega_i, a_i, t), \quad x_i^{u_i}(\omega_i, a_i, t) = y_i^{u_i}(\omega_i, a_i, t) - t(y_i^{u_i}(\omega_i, a_i, t)).$$

In an economy of $n \in \mathbb{N}$ individuals, net income distributions are given by

$$(x_1^{u_1}(\omega_1, a_1, t), \dots, x_n^{u_n}(\omega_n, a_n, t)),$$

parameterized by the wage vector $(a_1, \dots, a_n)$, endowment vector $(\omega_1, \dots, \omega_n)$, and tax schedule t . Each $x_i^{u_i}$ corresponds to individual i 's net income.

The income distribution in the absence of taxation is denoted by

$$(x_1^{u_1}(\omega_1, a_1, 0), \dots, x_n^{u_n}(\omega_n, a_n, 0)).$$

This baseline distribution serves as a reference point for assessing the impact of various tax policies on poverty.

4. Poverty-reducing tax schedules

This paper characterizes income tax schedules that guarantee robust poverty reduction across all possible economic environments. Specifically, we identify tax structures that decrease poverty—as measured by the poverty pre-order $\succsim_z$ defined in [Section 2](#)—regardless of the underlying distribution of endowments and wages, or the heterogeneity in individual preferences.

In formal terms, given a poverty pre-order $\succsim$, a tax schedule t is $\succsim$ -poverty-reducing if

$$(x_1^{u_1}(\omega_1, a_1, 0), \dots, x_n^{u_n}(\omega_n, a_n, 0)) \succsim (x_1^{u_1}(\omega_1, a_1, t), \dots, x_n^{u_n}(\omega_n, a_n, t))$$

for all endowment distributions $(\omega_1, \dots, \omega_n) \in \mathbb{R}_+^n$, all wage distributions $(a_1, \dots, a_n) \in \mathbb{R}_{++}^n$, all preference profiles $(u_1, \dots, u_n) \in \mathcal{U}^n$, and all income solution functions $x_1^{u_1}, \dots, x_n^{u_n}$.⁴

Our main result completely characterizes the tax schedules that reduce poverty uniformly across all possible wage, endowment, and preference distributions—the set of $\succsim_z$ -poverty-reducing schedules.

Theorem 1. *A tax schedule $t \in \mathcal{T}$ is $\succsim_z$ -poverty-reducing if and only if $t(y) = -s$ for all y with $y + s \leq z$ and some $s \geq 0$.*

This result states a tax schedule t is $\succsim_z$ -poverty-reducing if and only if it provides a fixed subsidy s to those individuals whose post-tax incomes fall below the poverty line z .

Remark 2. Without loss of generality, the subsidy parameter s in **Theorem 1** can be restricted to $s \in [0, z]$. When $s > z$, the condition $t(y) = -s$ for all y with $y + s \leq z$ is vacuously satisfied (the domain is empty) and carries no economic content. For any $\succsim_z$ -poverty-reducing schedule, the characterization holds with some $s \in [0, z]$: the case $s = 0$ corresponds to the zero-tax schedule (trivially poverty-reducing), while $s = z$ delivers a subsidy of z only at $y = 0$.

The formal proof of **Theorem 1** appears in **Appendix A**, with supporting intuition in the next subsection. **Section 4.2** discusses the scope and implications of **Theorem 1**.

4.1. Informal proof of Theorem 1

To provide intuition for the underlying logic, we offer a visual heuristic argument based on **Figures 1** to **3**.

Step 1 (Sufficiency): Fixed subsidies below z are poverty-reducing. Suppose $t \in \mathcal{T}$ satisfies $t(y) = -s$ for all y with $y + s \leq z$ and some $s \in [0, z]$. Fix arbitrary endowments, wages, preferences, and income solution functions, and let c and c' denote the corresponding pre- and post-tax income distributions censored at z .

To establish

$$(x_1^{u_1}(\omega_1, a_1, 0), \dots, x_n^{u_n}(\omega_n, a_n, 0)) \succsim_z (x_1^{u_1}(\omega_1, a_1, t), \dots, x_n^{u_n}(\omega_n, a_n, t)),$$

it suffices (by **Lemma 1** in **Appendix A**) to show $c'_i \geq c_i$ for each individual i .

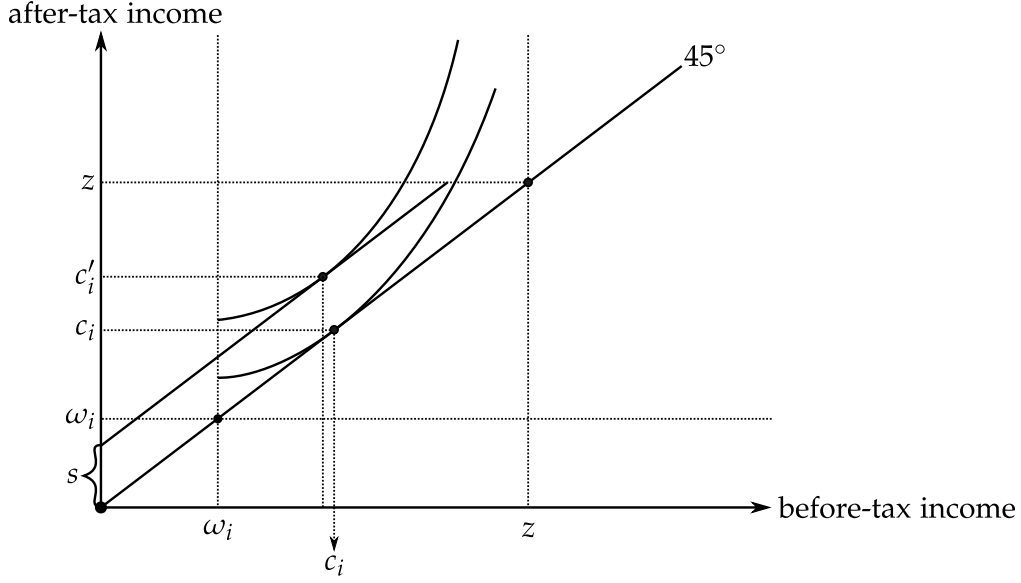
Fix i . In the absence of taxation, i solves

$$\max_{y_i \in [\omega_i, \omega_i + a_i]} u_i((y_i - \omega_i)/a_i, y_i).$$

As illustrated in **Figure 1**, since t acts as a pure subsidy for incomes below z , the budget line shifts upward by s . By normality of consumption, this generates a positive income effect, implying $c'_i \geq c_i$.

For non-poor incomes above z , after-tax incomes could fall below the poverty line—and hence violate $c'_i \geq c_i$ —only if the budget line were decreasing beyond z , which is ruled out by rank preservation since it would imply lower post-tax income at higher pre-tax income.

⁴In the absence of taxation, the solution functions are uniquely determined. However, multiple equilibrium solutions can emerge under taxation.

Figure 1: Individual i 's problem.

Step 2 (Necessity I): Poverty-reducing taxes must subsidize the poor. We now illustrate the necessity of subsidizing low incomes. Suppose that t is $\succsim_z$ -poverty-reducing. We claim that $t(y) \leq 0$ for all $y \in [0, z]$.

Proceeding by contradiction, suppose that $t(x^*) > 0$ for some $x^* \in [0, z]$. As illustrated in Figure 2, the corresponding net income $x^* - t(x^*)$ then lies strictly below the 45° line. Consider a homogeneous economy where all individuals have zero non-labor endowments, identical wages, and identical preferences, chosen such that each individual optimally selects income x^* in the absence of taxation. Let c and c' denote the resulting pre-tax and post-tax income distributions, respectively, each censored at z .

Since t is $\succsim_z$ -poverty-reducing, we must have $c' \succsim_D c$. In this homogeneous economy, this implies $c'_1 \geq c_1$. We now show this is impossible.

Under the tax t , individual 1's budget line must pass through the point $(x^*, x^* - t(x^*))$ and, by rank preservation, must otherwise lie within the shaded region of Figure 2. While Figure 2 illustrates a configuration consistent with the inequality $c'_1 \geq c_1$, the geometry necessarily implies a violation of the normality of consumption.

This illustrates why $t(x^*) > 0$ creates a tension: it forces consumption to be non-normal.

Step 3 (Necessity II): Subsidies below z must be uniform. We now complete the necessity argument. Suppose that t weakly subsidizes the poor but does not satisfy $t(y) = -s$ for all y with $y + s \leq z$.

Since t weakly subsidizes the poor, part of the budget constraint under t must lie above the 45° line with slope less than unity, as illustrated in Figure 3 by the segment through (y, c'_1) . Consider preferences satisfying consumption normality, as shown in Figure 3. Without taxation, the optimal income is $x^u(0, a, 0) = c_1$, while under t , the optimal after-tax income is $x^u(0, a, t) = c'_1 < c_1$. Constructing homogeneous distributions with these optimal

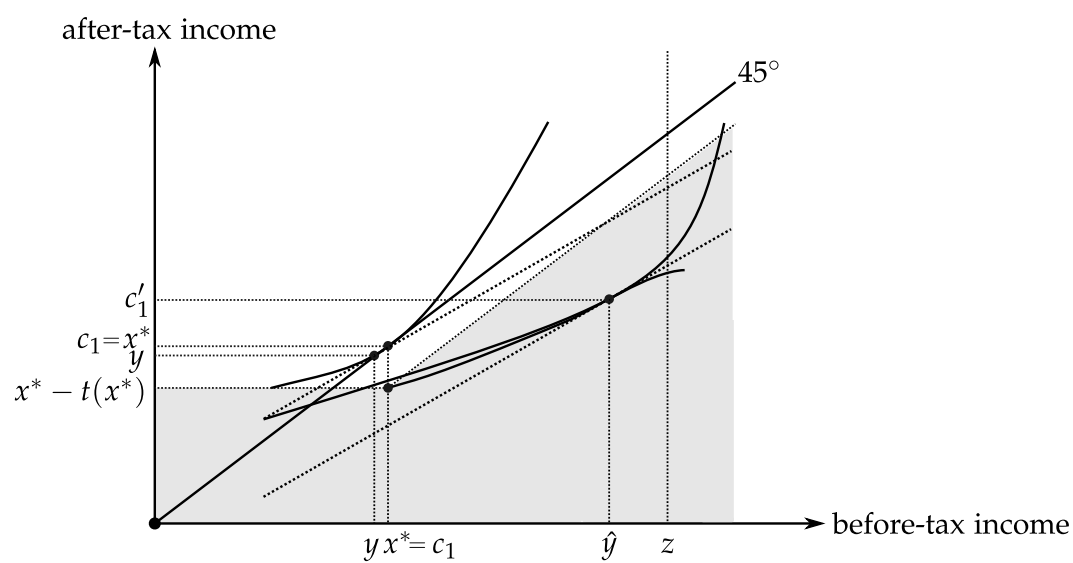


Figure 2: Individual 1's problem.

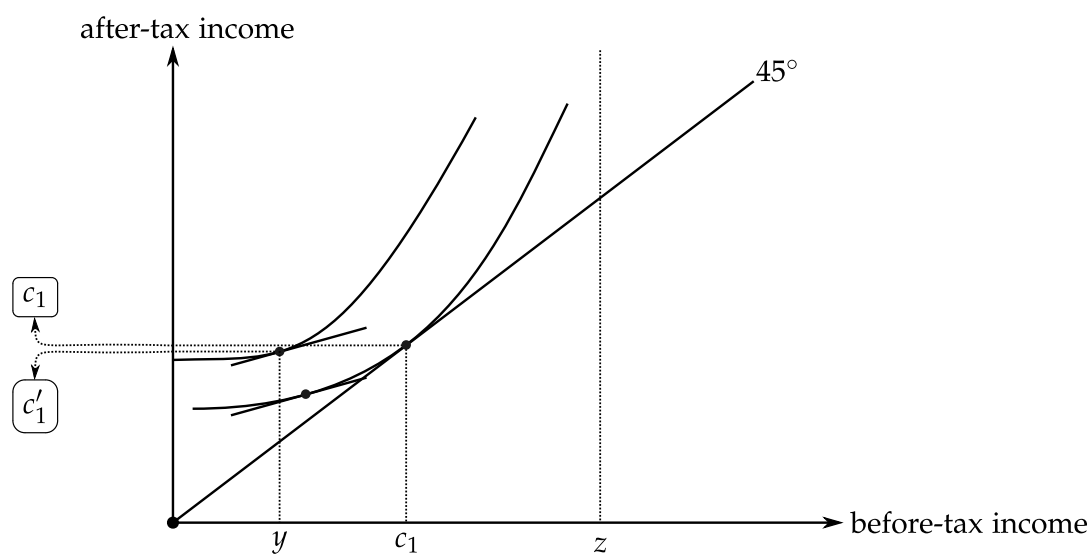


Figure 3: Individual 1's problem.

choices, where $z \geq c_1 > c'_1$, yields

$$(x^u(0, a, 0), \dots, x^u(0, a, 0)) \not\geq_z (x^u(0, a, t), \dots, x^u(0, a, t)),$$

contradicting that t is $\succsim_z$ -poverty-reducing. This illustrates why a uniform subsidy is necessary for all incomes remaining below z post-subsidy.

4.2. Scope, extensions, and robustness

This section examines the scope, economic implications, and robustness of our characterization result ([Theorem 1](#)). Our analysis proceeds in five strands.

First, we discuss the remarkable robustness of uniform lump-sum subsidies: they are independent of resource constraints, robust to arbitrary heterogeneity in preferences and endowments, and invariant across all poverty measures satisfying Focus, Symmetry, Continuity, and Monotonicity. Furthermore, the set of all poverty-reducing tax schedules is completely ordered by the magnitude of the lump-sum subsidy, facilitating comparisons across the policy-relevant space.

Second, we isolate the role of behavioral responses by comparing endogenous and exogenous income frameworks. Endogenizing labor supply transforms poverty alleviation from a mechanical redistribution problem—where any non-positive tax on the poor preserving non-poor status suffices—into one constrained by incentive compatibility, where only uniform lump-sum subsidies guarantee universal poverty reduction.

Third, we assess robustness to institutional complexities: endogenous poverty lines and multiple poverty thresholds.

Fourth, we assess robustness to alternative poverty criteria incorporating inequality among the poor.

Fifth, we contrast our outcome-based approach with complementary normative principles, yielding two fairness results: (i) an *equal benefit* characterization showing that uniform subsidies deliver identical utility gains to all remaining poor agents when preferences are quasilinear, along with a partial converse; and (ii) a result showing that *poverty preservation*—requiring that taxation not invert poverty rankings—implies poverty reduction but not conversely: uniform subsidies may violate preservation due to heterogeneous income effects. Finally, we contrast our framework with [Henry de Frahan and Maniquet’s \(2021\)](#) responsibility-sensitive welfarism, noting normative divergence but convergence in rejecting positive marginal tax rates on the poor.

Together, these extensions delineate the boundaries of our main result and clarify how endogenous labor supply reshapes the policy space for tax-mediated poverty alleviation.

4.2.1. Context-independence and preference-robustness

Our analysis notably omits explicit budgetary constraints on admissible tax schedules. Such constraints typically depend on contextual economic factors—specifically, the distribution of wage rates, the resulting gross income distribution, and the government’s resource capacity. Our analysis intentionally leaves these variables unspecified, as our universal poverty reduction framework applies across all economic environments.

Remarkably, this context-independence extends to preference specifications. Our characterization of poverty-reducing schedules imposes essentially no restrictions on

admissible preferences, a feature that sharply distinguishes it from related results in inequality-reducing taxation.

To understand this generality, consider the role of homogeneous societies—where all individuals share identical characteristics—in the proof structure. As the informal proof of [Theorem 1 \(Section 4.1\)](#) illustrates, homogeneous societies already impose the complete structure of poverty-reducing tax schedules: uniform subsidies below the poverty line. In such environments, the tax-free ordering of incomes is trivially preserved after taxation regardless of preferences, allowing us to isolate the essential features of poverty-reducing schedules without any preference restrictions. The necessity argument in [Appendix A](#) leverages this insight by constructing homogeneous counterexamples using specific functional forms—a Cobb-Douglas variant in [Lemma 2](#) and a convex combination of quasilinear and strictly concave leisure preferences in the main proof. These counterexamples demonstrate that any tax schedule violating the characterized structure fails to reduce poverty in at least one environment. The theorem does not require a large universe of preferences; rather, it permits essentially any preference space $\mathcal{U}^n$ containing these two standard functional forms.

This domain robustness is absent from the literature on inequality-reducing taxation (see, e.g., [Carbonell-Nicolau, 2025](#)), where homogeneous societies impose no meaningful structure—any tax leaves an already-equal distribution unchanged. There, the characterization must rely on heterogeneous environments where preserving the rank ordering between tax-free and post-tax distributions becomes critical, demanding suitable restrictions on admissible preferences.

4.2.2. Complete ordering of poverty-reducing tax schedules

Beyond the preference-independence property, our framework yields another significant finding: the set of universally poverty-reducing tax schedules exhibits a complete ordering under our poverty measure. This ordering naturally corresponds to the subsidy magnitude, s . More precisely, let $\mathcal{T}_{\succsim_z}$ be the set of all $\succsim_z$ -poverty-reducing tax schedules characterized by [Theorem 1](#):

$$\mathcal{T}_{\succsim_z} = \{t_s \in \mathcal{T} : t_s = -s \text{ for all } y \text{ with } y + s \leq z \text{ and some } s \in [0, z]\}.$$

For any two schedules $t_s, t_{s'} \in \mathcal{T}_{\succsim_z}$, a higher subsidy corresponds to a less impoverished distribution. Formally, $s \geq s'$ if and only if

$$(x_1^{u_1}(\omega_1, a_1, t_{s'}), \dots, x_n^{u_n}(\omega_n, a_n, t_{s'})) \succsim_z (x_1^{u_1}(\omega_1, a_1, t_s), \dots, x_n^{u_n}(\omega_n, a_n, t_s)) \quad (2)$$

for all endowment distributions $(\omega_1, \dots, \omega_n) \in \mathbb{R}_+^n$, all wage distributions $(a_1, \dots, a_n) \in \mathbb{R}_{++}^n$, all preference profiles $(u_1, \dots, u_n) \in \mathcal{U}^n$, and all income solution functions $x_1^{u_1}, \dots, x_n^{u_n}$.

This follows by examining the censored values x_s^z and $x_{s'}^z$ of each of i 's coordinates for the vectors in equation (2). By [Lemma 1](#) in [Appendix A](#), it suffices to show that these values satisfy

$$x_s^z \geq x_{s'}^z. \quad (3)$$

If $x_s^z = z$, inequality (3) holds trivially. Thus, consider the case where $x_s^z < z$.

In this case, consumer i 's gross income level y_s solves:

$$\max_{y \in [0, a_i]} u_i(y/a_i, y + s) = \max_{y \in [0, a_i]} u_i(y/a_i, y - t_s(y))$$

Since consumption is a normal good and $s \geq s'$, the solution $y_{s'}$ to

$$\max_{y \in [0, a_i]} u_i(y/a_i, y + s') \quad (4)$$

must satisfy

$$y_{s'} + s' \leq y_s + s = y_s - t_s(y_s) = x_s^z < z \quad (5)$$

Now observe that consumer i 's feasible set of gross income-net income bundles in (4) contains the set of bundles attainable under the problem

$$\max_{y \in [0, a_i]} u_i(y/a_i, y - t_{s'}(y)), \quad (6)$$

since $t_{s'}$ may impose non-zero marginal tax rates for gross incomes in the range $(z - s', +\infty)$.

Take any solution $y_{s'}$ to (6). Note that (5) implies $y_{s'} \leq z - s'$. Consequently, $(y_{s'}, y_{s'} + s')$ is also attainable in (6). Since the feasible set in (4) contains that in (6) and $y_{s'}$ maximizes utility over the larger set, it must also maximize utility over the smaller feasible set, implying $y_{s'} = y_s$ and

$$x_{s'}^z = y_{s'} - t_{s'}(y_{s'}) = y_{s'} + s' \leq y_s - t_s(y_s) = x_s^z,$$

Thus, (3) holds.

Conversely, suppose $s < s'$. Since $\mathcal{U}$ contains normal preferences (e.g., Cobb–Douglas), consider a single agent with zero endowment, sufficiently small wage $a_1 > 0$, and such preferences. Because $s, s' \in [0, z]$, choosing a_1 small enough ensures the agent remains poor under both subsidies: $y_s + s < z$ and $y_{s'} + s' < z$.

Since both budget constraints have slope 1 over the relevant domain, the comparison involves a pure income effect. Normality then implies an increase in net income:

$$y_{s'} + s' > y_s + s.$$

Consequently $x_{s'}^z > x_s^z$, violating the poverty ordering (2). Hence universal poverty ordering implies $s \geq s'$.

Combining both directions yields the equivalence.

4.2.3. Scope: equivalence across poverty measures

To clarify the scope of **Theorem 1**, we discuss the specific poverty measures it encompasses.

First, recall that for any two income distributions x and y , the relation $x \succsim_z y$ holds if and only if x exhibits no less poverty than y under all poverty measures that satisfy the axioms of Focus, Symmetry, Continuity, and Monotonicity.

This result implies that if a tax schedule t is $\succsim_z$ -poverty-reducing, then it is also poverty-reducing according to *any* poverty measure that satisfies these four axioms.

The converse assertion is also true. To see this, observe that any poverty pre-order $\succsim$ satisfying the four axioms is an extension of $\succsim_z$. That is, $\succsim \supseteq \succsim_z$ and $\succ \supseteq \succ_z$.

Now suppose t is $\succsim$ -poverty-reducing. Then, for any endowment distribution $(\omega_1, \dots, \omega_n) \in \mathbb{R}_+^n$, wage distribution $(a_1, \dots, a_n) \in \mathbb{R}_{++}^n$, preference profile $(u_1, \dots, u_n) \in \mathcal{U}^n$, and income solution functions $x_1^{u_1}, \dots, x_n^{u_n}$, we have

$$(x_1^{u_1}(\omega_1, a_1, 0), \dots, x_n^{u_n}(\omega_n, a_n, 0)) \succsim (x_1^{u_1}(\omega_1, a_1, t), \dots, x_n^{u_n}(\omega_n, a_n, t)).$$

Fix $(\omega_1, \dots, \omega_n), (a_1, \dots, a_n), (u_1, \dots, u_n)$, and $x_1^{u_1}, \dots, x_n^{u_n}$. Then, for each individual i ,

$$(x_i^{u_i}(\omega_i, a_i, 0), \dots, x_i^{u_i}(\omega_i, a_i, 0)) \succsim (x_i^{u_i}(\omega_i, a_i, t), \dots, x_i^{u_i}(\omega_i, a_i, t)). \quad (7)$$

This implies

$$x_i^{u_i}(\omega_i, a_i, t) \geq x_i^{u_i}(\omega_i, a_i, 0). \quad (8)$$

If this were not the case, we would have

$$(x_i^{u_i}(\omega_i, a_i, t), \dots, x_i^{u_i}(\omega_i, a_i, t)) \succ_z (x_i^{u_i}(\omega_i, a_i, 0), \dots, x_i^{u_i}(\omega_i, a_i, 0)),$$

which would further imply

$$(x_i^{u_i}(\omega_i, a_i, t), \dots, x_i^{u_i}(\omega_i, a_i, t)) \succ (x_i^{u_i}(\omega_i, a_i, 0), \dots, x_i^{u_i}(\omega_i, a_i, 0))$$

(since $\succsim$ is an extension of $\succsim_z$), contradicting (7).

Since (8) holds, **Lemma 1** in **Appendix A** implies

$$(x_1^{u_1}(\omega_1, a_1, 0), \dots, x_n^{u_n}(\omega_n, a_n, 0)) \succsim_z (x_1^{u_1}(\omega_1, a_1, t), \dots, x_n^{u_n}(\omega_n, a_n, t)).$$

Thus, t is also $\succsim_z$ -poverty-reducing.

These insights are formalized in the following corollary to **Theorem 1**:

Corollary 1 (to **Theorem 1**). *Suppose $\succsim$ is an extension of the core pre-order $\succsim_z$, i.e., $\succsim \supseteq \succsim_z$ and $\succ \supseteq \succ_z$. A tax schedule $t \in \mathcal{T}$ is $\succsim$ -poverty-reducing if and only if $t(y) = -s$ for all y with $y + s \leq z$ and some $s \geq 0$.*

The class of poverty measures covered by this corollary includes most well-known indices used in applied and theoretical work. In particular, standard members of the Foster–Greer–Thorbecke (FGT) family (**Foster et al., 1984**) with parameter $\alpha > 0$, the Atkinson-style poverty indices (**Atkinson, 1987**), the Watts index (**Watts, 1968**), and the Sen–Shorrocks–Thon (**Thon, 1979; Shorrocks, 1995**) family all satisfy the axioms of Focus, Symmetry, Continuity, and Monotonicity. Consequently, each of these measures induces a poverty ordering that extends the core pre-order $\succsim_z$.

Thus, for all such indices, the concept of poverty reduction is equivalent.

Remark 3. Some poverty indices, such as certain members of the FGT family, rank a policy that lifts many to the poverty line but slightly reduces the income of the poorest as poverty-reducing. The core partial order $\succsim_z$, however, leaves these two distributions incomparable, as no single axiom—Focus, Symmetry, Continuity, or Monotonicity—applies to such a multidirectional change.

This apparent divergence between a completion and the underlying partial order does not undermine **Theorem 1** and **Corollary 1**. Taken together, these results assert that a tax schedule reduces poverty under $\succsim_z$ if and only if it reduces poverty under *any* poverty measure satisfying the four axioms—that is, under any extension of $\succsim_z$. This equivalence

holds because the paper's notion of poverty reduction is universal: a tax must reduce poverty in every economic environment. Any tax that lowers the poorest type's income in some environment fails this test: in a homogeneous economy populated solely by that type, the tax would reduce everyone's income, increasing poverty under every admissible measure. Such policies are therefore correctly excluded.

In short, the framework deliberately rules out policies that trade off the welfare of some against gains for others.

4.2.4. Extension to non-monotonic tax schedules

In [Appendix B](#), we characterize poverty-reducing tax schedules when the monotonicity constraint is relaxed, permitting negative marginal tax rates. This extension is particularly relevant given findings in the optimal taxation literature that negative marginal rates can be optimal for low-income individuals (see, e.g., [Kanbur et al., 1994](#); [Maniquet and Neumann, 2021](#); [Henry de Frahan and Maniquet, 2021](#)).

For this extension, we redefine a tax schedule as a continuous and rank-preserving function $t : \mathbb{R}_+ \rightarrow \mathbb{R}$ satisfying $t(y) \leq y$ for all $y \geq 0$, dropping the monotonicity requirement. Let $\mathcal{T}^*$ denote this expanded set of admissible schedules.

Theorem 2. *A tax schedule $t \in \mathcal{T}^*$ is $\succsim_z$ -poverty-reducing if and only if $t(y) \leq 0$ for all $y \in [0, z]$ and t is non-increasing over the set $[0, \max\{y \in [0, z] : y - t(y) \leq z\}]$.*

The formal proof is provided in [Appendix B](#). While pure subsidies ([Theorem 1](#)) are no longer the unique solution, two core insights remain: (i) subsidization of the poor is necessary, and (ii) positive marginal tax rates on the poor remain incompatible with poverty reduction—though marginal rates may now be negative rather than merely zero. Thus, even when we expand the feasible policy set to include non-monotonic schedules, our findings remain qualitatively similar, with negative marginal tax rates emerging as a permissible tool for creating work incentives while subsidizing the poor.

4.2.5. Alternative poverty criteria

The main analysis employs the censored first-order stochastic dominance (FOSD) criterion $\succsim_z$. However, recent work has developed distribution-sensitive poverty measures that explicitly account for inequality among the poor. Following [Carbonell-Nicolau \(2026\)](#), we consider two such criteria: the relative poverty pre-order $\succsim_R^z$ and its absolute counterpart $\succsim_A^z$. These measures satisfy the standard axioms of Focus, Symmetry, and Continuity, but replace the Monotonicity axiom with distribution-sensitivity axioms. Specifically, they combine the Minimal Transfer axiom with either Relative Weak Monotonicity (for $\succsim_R^z$) or Absolute Weak Monotonicity (for $\succsim_A^z$), as detailed in [Carbonell-Nicolau \(2026\)](#).

Intuitively, these criteria evaluate poverty through inequality-adjusted measures introduced by [Shorrocks \(1983\)](#), applied to censored income distributions. Under $\succsim_R^z$, a distribution y exhibits less relative poverty than x if the censored distribution y^z is at least as equal as x^z in relative terms (via the relative Lorenz criterion) *and* has weakly higher total income. Analogously, $\succsim_A^z$ applies the absolute Lorenz criterion to censored distributions, requiring that absolute income gaps among the poor do not widen while total income does not decline. Both measures reject income increases that exacerbate inequality among the poor—even when all poor individuals become richer in absolute terms. In

contrast, censored FOSD would deem such changes unambiguously poverty-reducing. As demonstrated in [Carbonell-Nicolau \(2026\)](#), these criteria are non-nested with censored FOSD: neither implies the other in general.

A natural question is whether [Theorem 1](#) extends to these distribution-sensitive poverty criteria. The answer is partial. The necessity direction carries over completely:

Theorem 3. *If a tax schedule $t \in \mathcal{T}$ is $\succsim_R^z$ -poverty-reducing (respectively, $\succsim_A^z$ -poverty-reducing), then $t(y) = -s$ for all y with $y + s \leq z$ and some $s \geq 0$.*

This result follows directly from the necessity portion of [Theorem 1](#)'s informal argument (see [Section 4.1](#)), which relies entirely on homogeneous economies—environments where all individuals share identical endowments, wages, and preferences. In such settings, the three poverty criteria ($\succsim^z$, $\succsim_R^z$, and $\succsim_A^z$) coincide: with no heterogeneity among agents, inequality among the poor cannot arise or change, rendering the distribution-sensitivity of $\succsim_R^z$ and $\succsim_A^z$ vacuous. Consequently, any tax schedule violating the fixed-subsidy structure increases poverty in some homogeneous economy under all three criteria, establishing the necessity direction for the alternative measures.

The sufficiency direction, however, fails for both $\succsim_R^z$ and $\succsim_A^z$. To see why, consider a fixed lump-sum subsidy $t(y) = -s$ for $y + s \leq z$ with $s > 0$, applied to a two-agent economy where both agents remain poor both before and after taxation. Suppose agent preferences (or wage rates) are such that the subsidy induces a 10% consumption increase for the poorer agent but a 15% increase for the richer poor agent.

Such heterogeneous responses are entirely plausible under our framework, which allows arbitrary preference heterogeneity (and can even arise with common preferences given sufficient wage dispersion). While both agents become absolutely better off, relative inequality among the poor increases. Under $\succsim_R^z$, the post-tax censored distribution neither dominates nor is dominated by the tax-free distribution. An analogous construction using absolute (rather than proportional) consumption gains demonstrates the failure of sufficiency for $\succsim_A^z$.

This incomparability implies that no tax schedule can universally reduce poverty under $\succsim_R^z$ or $\succsim_A^z$ across all preference profiles. Any candidate will generate some environment where behavioral responses worsen inequality among the poor. However, the literature on tax-mediated inequality and polarization (see, e.g., [Carbonell-Nicolau and Llavador, 2018, 2021a,b, 2025](#)) suggests that under the common assumption of homogeneous preferences, it is possible to derive conditions on wage elasticities such that uniform lump-sum subsidies (which are marginal-rate progressive) are poverty-reducing with respect to $\succsim_R^z$ and $\succsim_A^z$.

4.2.6. On the role of rank preservation

The rank preservation requirement—that post-tax income $y - t(y)$ be non-decreasing in pre-tax income y —plays an interesting dual role in our analysis. Geometrically, a violation occurs precisely when the marginal tax rate exceeds 100%, creating a downward-sloping segment in the (y, x) diagram where higher pre-tax income yields lower post-tax consumption. Such segments are economically irrelevant for rational agents: if $y_2 > y_1$ but $y_2 - t(y_2) < y_1 - t(y_1)$, then working less (earning y_1) strictly dominates working more (earning y_2). No utility-maximizing individual would ever select a point on a downward-sloping portion of the budget constraint.

This observation raises a natural question: is rank preservation mathematically necessary for our characterization results? The answer is no. The proof of [Theorem 1](#) relies only on behavior at points that rational agents might actually choose—the non-dominated “upper envelope” of the budget constraint where $x(y)$ is non-decreasing. For any tax schedule violating rank preservation, we can construct its upper envelope

$$\tilde{x}(y) = \max_{y' \leq y} y' - t(y'),$$

which satisfies rank preservation by construction. Since agents optimize over the upper envelope regardless of the underlying schedule’s shape, the necessity arguments in Steps 2 and 3 of the informal proof of [Theorem 1](#) ([Section 4.1](#)) apply almost verbatim to $\tilde{x}(y)$. Consequently, the characterization in [Theorem 1](#) holds for the economically relevant portion of the budget constraint.

Nevertheless, we maintain rank preservation as a modeling assumption because it serves important economic purposes beyond mathematical necessity. Normatively, it embodies a minimal horizontal equity principle—taxation should not reverse income rankings—widely regarded as a baseline fairness requirement. Practically, a violation would create perverse incentives for workers to reduce output, generating inefficient bunching at kink points. Analytically, it ensures tractability by keeping the entire budget constraint potentially relevant for optimization, avoiding the need to distinguish between dominated and non-dominated portions.

4.2.7. Robustness across multiple poverty lines

Policymakers often debate the appropriate poverty threshold, raising a natural robustness question: does a tax policy reduce poverty regardless of where the cutoff is set within a specified range $Z \subseteq \mathbb{R}_{++}$? Requiring poverty reduction for *all* $z \in Z$ —rather than a single arbitrary threshold—provides a more compelling basis for policy evaluation.

[Theorem 1](#) extends naturally to this multi-threshold setting. We seek to characterize tax schedules that are $\succsim_z$ -poverty-reducing for *every* $z \in Z$. The characterization reduces to a simple observation: it suffices to consider only $\sup Z$.

Sufficiency. Suppose t satisfies $t(y) = -s$ for all y with $y + s \leq \sup Z$ and some $s \geq 0$. Fix any $z \in Z$. Since $z \leq \sup Z$, we have

$$\{y : y + s \leq z\} \subseteq \{y : y + s \leq \sup Z\}.$$

Thus $t(y) = -s$ for all y with $y + s \leq z$, so by [Theorem 1](#), t is $\succsim_z$ -poverty-reducing.

Necessity. Suppose t is $\succsim_z$ -poverty-reducing for all $z \in Z$. If $\sup Z \in Z$, then t is $\succsim_{\sup Z}$ -poverty-reducing, so [Theorem 1](#) immediately yields $t(y) = -s$ for all y with $y + s \leq \sup Z$.

If $\sup Z \notin Z$, choose a sequence $(z_k) \subseteq Z$ with $z_k \rightarrow \sup Z$. For each k , [Theorem 1](#) gives $s_k \in [0, z_k]$ such that $t(y) = -s_k$ whenever $y + s_k \leq z_k$. Since $0 + s_k \leq z_k$ for all k , we have $t(0) = -s_k$ for all k , so the sequence (s_k) is constant: $s_k = s = -t(0)$.

Now fix y with $y + s \leq \sup Z$. If $y + s < \sup Z$, then $y + s \leq z_k$ for all sufficiently large k , implying $t(y) = -s$. If $y + s = \sup Z$ and $y > 0$, take a sequence $y_m \uparrow y$; then $y_m + s < \sup Z$ for all m , so $t(y_m) = -s$ and continuity of t yields $t(y) = -s$. The case $y = 0$ holds by construction since $t(0) = -s$.

Thus, robustness across an entire set of poverty lines imposes no additional structural restrictions beyond those in [Theorem 1](#)—it merely requires that the subsidy domain be defined relative to $\sup Z$.

Corollary 2 (to [Theorem 1](#)). *Let $Z \subseteq \mathbb{R}_{++}$ be nonempty. A tax schedule $t \in \mathcal{T}$ is $\succsim_z$ -poverty-reducing for all $z \in Z$ if and only if $t(y) = -s$ for all y with $y + s \leq \sup Z$ and some $s \geq 0$.*

This characterization holds for *any* nonempty set Z —boundedness affects only whether the subsidy domain is finite or universal, not the underlying structural form.

4.2.8. Endogenous poverty lines

Throughout the paper the poverty line $z > 0$ is treated as exogenously given. In practice, however, many official thresholds are defined relative to the income distribution itself—for example as a fixed percentage of median income. This raises the question of whether [Theorem 1](#) extends to such endogenous poverty lines.

The answer depends on how the poverty line is determined. When the threshold is defined using a *post-tax* statistic (e.g., a fraction of the post-tax median), a circularity problem arises: the poverty line z is itself an equilibrium object that depends on the tax schedule t under consideration. Evaluating policy relative to z therefore requires solving a fixed-point problem (t, z) , in which each variable depends on the other. This circularity constitutes a fundamental obstacle that we do not address in this paper.

Circularity can be avoided by defining the poverty line using a *pre-determined* statistic—such as a lagged median or any function of structural primitives. In such cases, the threshold becomes a well-defined mapping $z = z(\omega', a', u')$, where (ω', a', u') serve as reference parameters that fix the poverty line prior to policy implementation (e.g., last year’s economic fundamentals).⁵

This raises a natural question: does there exist a single tax schedule t satisfying

$$(x_1^{u_1}(\omega_1, a_1, 0), \dots, x_n^{u_n}(\omega_n, a_n, 0)) \succsim_{z(\omega', a', u')} (x_1^{u_1}(\omega_1, a_1, t), \dots, x_n^{u_n}(\omega_n, a_n, t))$$

for *all* reference parameters (ω', a', u') and all actual economies (ω, a, u) ?

The answer is affirmative. Let

$$Z = \{z(\omega', a', u') : (\omega', a', u') \in \mathbb{R}_+^n \times \mathbb{R}_{++}^n \times \mathcal{U}^n\}$$

denote the range of the poverty-line function. By [Corollary 2](#), a single tax schedule t with $t(y) = -s$ for all y satisfying $y + s \leq \sup Z$ (and any fixed $s \geq 0$) is $\succsim_z$ -poverty-reducing for *every* $z \in Z$.

When $\sup Z < \infty$ —which holds naturally when poverty lines anchor to predetermined reference statistics drawn from a bounded domain of historical or projected fundamentals—the subsidy targets incomes below the finite threshold $\sup Z$.

Thus, [Corollary 2](#) implies that a single tax schedule robustly reduces poverty across all reference-dependent thresholds.

⁵For comparison, in environments with *exogenous* incomes, [Carbonell-Nicolau \(2026\)](#) handles endogenous poverty lines via normalization arguments (exploiting scale or translation invariance). Such approaches fail here because behavioral responses break scaling equivariance: with endogenous labor supply, proportional changes in primitives generally induce non-proportional income adjustments. We therefore adopt a direct characterization rather than pursuing a normalization strategy.

Corollary 3 (to **Theorem 1**). Let $z : \mathbb{R}_+^n \times \mathbb{R}_{++}^n \times \mathcal{U}^n \rightarrow \mathbb{R}_{++}$ be a poverty line function mapping reference parameters (ω', a', u') to thresholds $z(\omega', a', u') > 0$, and let Z denote its range. A tax schedule $t \in \mathcal{T}$ is $\succsim_z$ -poverty-reducing for all $z \in Z$ if and only if $t(y) = -s$ for all y with $y + s \leq \sup Z$ and some $s \geq 0$.

This characterization upholds the robustness requirement central to **Theorem 1**: poverty-reducing policies must work across all economic environments, not merely specific configurations of fundamentals. Even when calibrated to reference information—such as lagged income distributions—the actual implementation environment (wages, endowments, preferences) remains uncertain. The single-schedule solution in **Corollary 3** satisfies this requirement without requiring policymakers to adjust the tax code as reference statistics evolve.

4.2.9. Responsibility-sensitive justice versus outcome-based poverty reduction

The responsibility-sensitive framework, as formalized by [Fleurbaey and Maniquet \(2006, 2007\)](#) and recently applied to poverty reduction by [Henry de Frahan and Maniquet \(2021\)](#), distinguishes between inequalities arising from circumstances beyond individual control (e.g., wage rates) and those reflecting autonomous preference differences (e.g., taste for leisure). Their social welfare functions satisfy an equal-wage transfer axiom: absent informational constraints, no redistribution should occur among agents with identical wages but different labor supplies. This principle implies that deserved poverty—resulting from freely chosen low labor supply among agents with adequate earning capacity—should not be remedied through redistribution, whereas undeserved poverty stemming from low wages warrants intervention. Their optimal tax formula features zero or negative marginal rates on low incomes, followed by a discrete jump to positive rates just above the income level of the hardworking poor (those working full time at the minimum wage), with bunching at this threshold.

By contrast, our framework adopts a public-reason perspective in which income below a socially recognized threshold constitutes an objective deprivation demanding redress irrespective of its behavioral origin. This normative stance reflects taxpayers' concern with observable manifestations of poverty rather than unobservable leisure choices ([Besley and Coate, 1995](#)), and yields a uniform lump-sum subsidy for all individuals whose post-tax incomes fall below the poverty line—a policy robust across all preference profiles and economic environments.

Despite these normative differences, both frameworks converge on rejecting positive marginal tax rates for the poor. In our setting, such rates violate universal poverty reduction by lowering post-tax income below its tax-free level through adverse substitution effects. In the responsibility framework, they violate fairness by transferring resources from hardworking to less hardworking agents with identical wages.

4.2.10. Endogenous versus exogenous income

To isolate the role of behavioral responses in poverty-reducing taxation, we compare the endogenous income framework developed in this paper with its exogenous counterpart. This comparison requires adopting a common poverty criterion. While the exogenous income analysis in [Carbonell-Nicolau \(2026\)](#) employs distribution-sensitive poverty measures

non-nested with censored FOSD, we can adapt those methods to characterize poverty-reducing taxes under censored FOSD in the exogenous case, yielding a clean benchmark for comparison.

In the exogenous setting, a *tax schedule* is a continuous function $t: \mathbb{R}_+ \rightarrow \mathbb{R}$ satisfying $t(x) \leq x$ for all $x \geq 0$.

Given a poverty line $z > 0$, a tax schedule t is $\succsim_z$ -poverty-reducing if, for every pre-tax income distribution $\mathbf{x} = (x_1, \dots, x_n)$,

$$\mathbf{x} \succsim_z (x_1 - t(x_1), \dots, x_n - t(x_n)).$$

The following result characterizes poverty-reducing tax schedules when income is exogenous:

Proposition 1. *A tax schedule t is $\succsim_z$ -poverty-reducing if and only if it satisfies:*

- (i) *Non-poor status preservation:* $x > z \Rightarrow x - t(x) \geq z$;
- (ii) *Weak subsidization:* $t(x) \leq 0$ for all $x \in [0, z]$.

Condition (i) ensures that taxation does not push the initially non-poor into poverty—a basic fairness requirement. Condition (ii) guarantees that every poor individual receives a non-negative transfer.

The characterization in **Proposition 1** stands in contrast to the findings in **Carbonell-Nicolau (2026)**, where poverty-reducing tax schedules relative to distribution-sensitive poverty measures require additional structure: rank preservation among the poor and average-rate progressivity.

The sufficiency direction in **Proposition 1** is straightforward. Suppose t satisfies (i) and (ii). Let $\mathbf{x}$ be any pre-tax distribution and

$$\mathbf{y} = (x_1 - t(x_1), \dots, x_n - t(x_n))$$

the post-tax distribution. Censor both at z to obtain $\mathbf{x}^z$ and $\mathbf{y}^z$. For each individual i :

- If $x_i \leq z$, then $y_i = x_i - t(x_i) \geq x_i$ by (ii), so $y_i^z \geq x_i^z$.
- If $x_i > z$, then $y_i \geq z$ by (i), so $y_i^z = z = x_i^z$.

Thus $y_i^z \geq x_i^z$ for all i , which implies $\mathbf{x} \succsim_z \mathbf{y}$.

For necessity, suppose first that $t(x^*) > 0$ for some $x^* \in [0, z]$. Consider the homogeneous distribution $\mathbf{x} = (x^*, \dots, x^*)$. Then

$$\mathbf{y} = (x^* - t(x^*), \dots, x^* - t(x^*))$$

with $y_i < x_i \leq z$, so $\mathbf{y}^z = \mathbf{y}$ and $\mathbf{x}^z = \mathbf{x}$. Since $y_i < x_i$ for all i , we have $\mathbf{x} \not\succsim_z \mathbf{y}$, contradicting poverty reduction.

Second, suppose t violates non-poor status preservation: there exists $x' > z$ with $x' - t(x') < z$. Consider $\mathbf{x} = (x', \dots, x')$. Then $\mathbf{x}^z = (z, \dots, z)$ but

$$\mathbf{y}^z = (x' - t(x'), \dots, x' - t(x')),$$

with all components less than z , so again $\mathbf{x} \not\succsim_z \mathbf{y}$.

Hence both (i) and (ii) are necessary.

Under $\succsim_z$, only individual income levels matter—not their relative ordering or dispersion. This allows for tax schedules that violate rank preservation or monotonicity yet still reduce poverty.

Example 1 (Rank reversal without increasing poverty). Let $z = 10$, and consider two poor individuals with pre-tax incomes (4, 6). Define a tax schedule with subsidies $t(4) = -3$ and $t(6) = -1$, so post-tax incomes are (7, 5). Ranks are reversed, but both individuals are better off. Thus poverty is reduced despite rank reversal.

Example 2 (Non-monotonic subsidies). Let $z = 10$, and consider two individuals with pre-tax incomes (3, 5). Define a tax schedule with $t(3) = -1$ and $t(5) = -3$, so that tax liability decreases as income increases—violating monotonicity. Nevertheless, post-tax incomes are (4, 8), which dominate the pre-tax vector (3, 5) pointwise.

The divergence between **Proposition 1** and **Theorem 1** is driven by behavioral responses. With exogenous income, any non-positive tax on $[0, z]$ that preserves non-poor status mechanically raises their income. With endogenous income, however, declining marginal subsidies may cause individuals to work less, with post-subsidy income falling below its tax-free level.

Only a uniform lump-sum subsidy eliminates these substitution effects. Its zero marginal tax rates generate a pure income effect, unambiguously raising consumption for all poor individuals. Thus, endogenizing income transforms poverty reduction from a mechanical problem into one constrained by incentive compatibility.

4.2.11. Equal benefit and poverty-reducing subsidies

We adapt Young's (1987; 1988; 1990) equal sacrifice principle to the context of poverty-reducing subsidies. While Young's framework focuses on the equitable distribution of utility *losses* through taxation, our setting considers utility *gains* accruing from transfers to the poor. Restricting attention to the family of poverty-reducing schedules characterized in **Theorem 1**, we ask which policies deliver *equal benefit* to those who remain poor after subsidization.

This criterion stipulates that any individual remaining below the poverty line post-subsidy must experience an identical welfare improvement. We demonstrate that poverty-reducing schedules satisfy equal benefit if preferences are quasilinear. Moreover, we establish a partial converse: the absence of a kink at $z - s$ and the elimination of income effects on labor supply are both necessary for equal benefit among the poor.

Definition 1. A tax schedule $t \in \mathcal{T}$ satisfies *equal benefit among the poor with respect to* $u \in \mathcal{U}$ if there exists a constant $\alpha \geq 0$ such that

$$u(l^u(\omega, a, t), x^u(\omega, a, t)) - u(l^u(\omega, a, 0), x^u(\omega, a, 0)) = \alpha$$

for all (ω, a) such that $x^u(\omega, a, t) \leq z$.

A utility function $u \in \mathcal{U}$ is *quasilinear* if it takes the form $u(l, x) = v(l) + x$, where $v : [0, 1] \rightarrow \mathbb{R}$ is continuous, strictly decreasing, strictly concave, and differentiable on $(0, 1)$. Under quasilinearity, the marginal utility of consumption is constant, eliminating income effects on labor supply while maintaining normality of consumption.

Recall from **Theorem 1** that $\succsim_z$ -poverty-reducing schedules take the form $t_s(y) = -s$ for all y satisfying $y + s \leq z$ and some $s \in [0, z]$. The following proposition identifies conditions under which such schedules implement equal benefit among the poor.

Proposition 2. *A member t_s of the family of $\succsim_z$ -poverty-reducing tax schedules implements equal benefit among the poor with respect to $u \in \mathcal{U}$ if u is quasilinear and t_s has no kink point at $z - s$.*

The intuition is straightforward: equal benefit requires that all agents remaining poor post-subsidy experience an identical utility gain. Quasilinear preferences ensure this by eliminating income effects on labor supply—so a lump-sum transfer leaves hours worked unchanged—and by making utility affine in consumption. Consequently, a uniform subsidy s translates into a uniform utility gain of s for all recipients. The no-kink condition at $z - s$ ensures the subsidy remains uniformly s throughout the domain of agents who stay poor.

Suppose $u(l, x) = v(l) + x$ is quasilinear and $t_s(y) = -s$ for all $y \in [0, z - s]$, with no kink at $z - s$. Consider an agent with parameters (ω, a) whose gross income under the subsidy satisfies $y^u(\omega, a, t_s) \leq z - s$. The absence of a kink at $z - s$ ensures that the agent faces a uniform subsidy of s over the relevant domain, so their labor supply solves

$$\max_{l \in [0, 1]} v(l) + \omega + al + s.$$

Since s enters additively, this problem has the same solution as the tax-free problem

$$\max_{l \in [0, 1]} v(l) + \omega + al,$$

so labor supply remains unchanged: $l^u(\omega, a, t_s) = l^u(\omega, a, 0)$.

Observe that only agents with $y^u(\omega, a, t_s) \leq z - s$ remain poor post-subsidy. To see this, note that when $y = y^u(\omega, a, t_s) > z - s$, rank preservation of t_s and the no-kink condition ensure that net income exceeds the poverty line beyond this point, yielding $x^u(\omega, a, t_s) > z$.

For the agents who remain poor after the subsidy, the utility gain equals

$$[v(l^u(\omega, a, t_s)) + x^u(\omega, a, t_s)] - [v(l^u(\omega, a, 0)) + x^u(\omega, a, 0)] = s,$$

which equals s for all types (ω, a) —hence equal benefit holds with $\alpha = s$.

Having established sufficient conditions for equal benefit, we now identify two necessary properties: (i) the absence of a kink at $z - s$, and (ii) the elimination of income effects on labor supply for all agents remaining poor.

Proposition 3. *If t_s implements equal benefit among the poor with respect to $u \in \mathcal{U}$, then the following must hold:*

- (i) t_s has no kink point at $z - s$;
- (ii) For every (ω, a) such that $(y^u(\omega, a, t_s), x^u(\omega, a, t_s)) = (y, y + s)$ with $y \leq z - s$, the removal of subsidy s generates no income effect on labor supply.

To see why the no-kink condition is necessary, suppose first that t_s has a kink point at $z - s$. Then there exists $a > 0$ and an open interval V_a containing a such that

$$y^u(z - s, a', t_s) = z - s, \quad \text{for all } a' \in V_a, \quad \text{and} \quad y^u(z - s, a, 0) > z - s$$

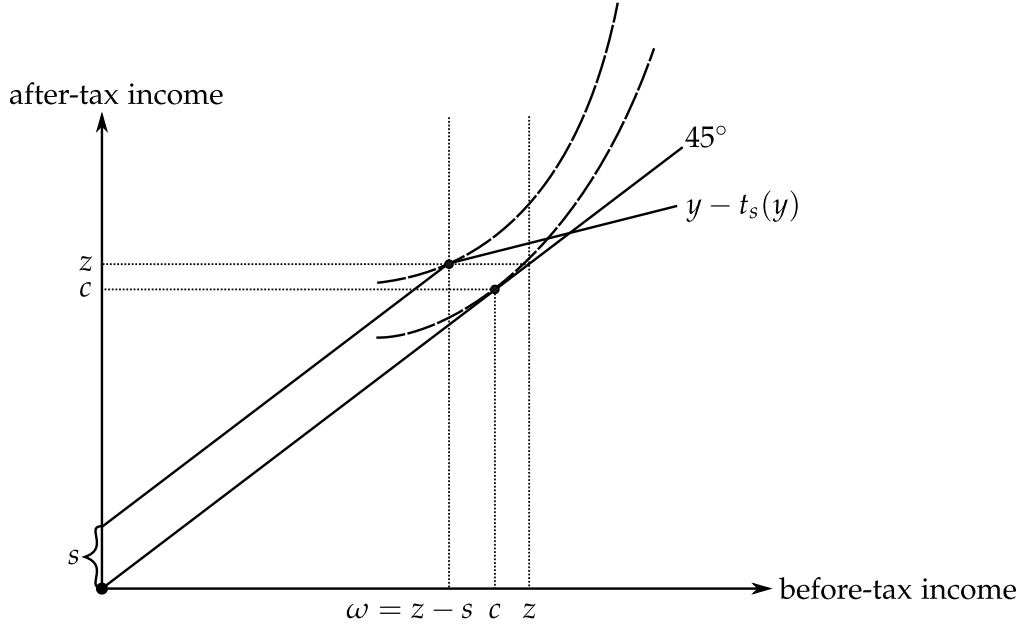


Figure 4: Endowment $\omega = z - s$ and wage a ; tax-free income $c = y^u(\omega, a, 0)$; wage increase raises tax-free utility but not post-subsidy utility.

(Figure 4).⁶ Then $x^u(z - s, a, t_s) = z$, and the benefit from the subsidy at $(z - s, a)$ is given by

$$u(0, z) - u(l^u(z - s, a, 0), x^u(z - s, a, 0)).$$

Now slightly increase the wage to a' within V_a . In the tax-free regime, utility increases as the higher wage expands the agent's feasible set. Post-subsidy, however, the agent remains at the kink point $(z - s, z)$: by [Mirrlees's \(1971\) agent monotonicity condition](#)—which is satisfied when consumption is a normal good (see [Myles, 1995](#), p. 136)—the indifference curve through $(z - s, z)$ becomes steeper as the wage increases, ensuring the kink remains optimal for sufficiently small wage increments. Consequently, post-subsidy utility is locally

⁶Under schedule t_s with a kink at $z - s$, the budget constraint in a neighborhood B of $z - s$ is

$$x = \begin{cases} y + s & \text{if } y \leq z - s, \\ z + (1 - \tau)(y - (z - s)) & \text{if } B \ni y > z - s, \end{cases}$$

where $\tau \in (0, 1]$ is the marginal tax rate just above the kink (phase-out rate of the subsidy). Since $\omega = z - s$ is the minimum feasible gross income, the agent's choice set begins precisely at the kink point $(z - s, z)$, and for small enough $y > z - s$ the budget line has slope $1 - \tau < 1$.

The conditions in (1) ([Footnote 3](#)) imply that for every (y, x) and ω ,

$$\lim_{a \rightarrow \infty} MRS_{(\omega, a)}^u(y, x) = -(1/a) \cdot \frac{\partial u((y - \omega)/a, x)}{\partial l} \bigg/ \frac{\partial u((y - \omega)/a, x)}{\partial x} = 0 \quad \text{and} \quad \lim_{a \downarrow y} MRS_{(\omega, a)}^u(y, x) = +\infty.$$

Since the map $a \mapsto MRS_{(\omega, a)}^u(y, x)$ is continuous for every (y, x) and ω , the Intermediate Value Theorem gives $a^* > 0$ such that $MRS_{(z-s, a^*)}^u(z - s, z) = 1$. Consequently, $y^u(z - s, a^*, t_s) = z - s$. Moreover, removing the lump-sum subsidy s generates a pure income effect, and since leisure is not inferior, the agent with wage a^* supplies the same or more labor: $y^u(z - s, a^*, 0) \geq z - s$. Now increasing the wage rate slightly from a^* to $\tilde{a} > a^*$ does not change the post-subsidy bundle since that is a kink point, whereas it strictly increases $y^u(z - s, \tilde{a}, 0)$ above $z - s$.

invariant to the wage, while tax-free utility strictly increases—reducing the net benefit and violating the equal benefit criterion.

Thus if t_s implements equal benefit among the poor, there is no kink point at $z - s$.

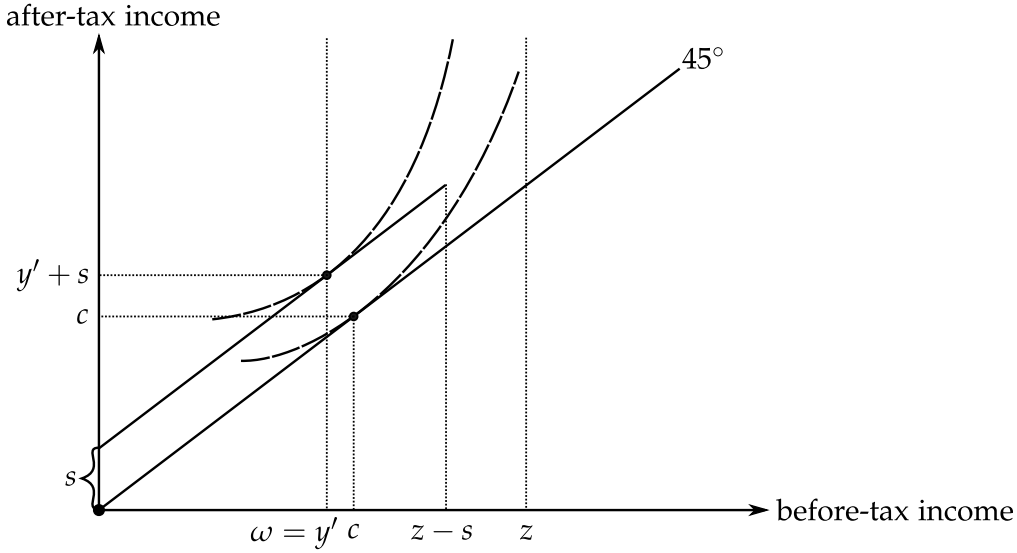


Figure 5: Endowment $\omega = y'$ and wage a' ; tax-free income $c = y^u(y', a', 0)$; income effect violation: wage decrease lowers tax-free utility but not post-subsidy utility

To understand the necessity of eliminating income effects, fix a consumption bundle $(y', y' + s)$ with $y' \leq z - s$. Any agent choosing this bundle remains poor after the subsidy. Suppose that $(y', y' + s)$ is chosen post-subsidy by an agent with endowment ω_0 and wage a :

$$(y', y' + s) = (y^u(\omega_0, a, t_s), x^u(\omega_0, a, t_s)).$$

Then $MRS_{(\omega_0, a)}^u(y', y' + s) \geq 1$, with strict inequality possible at corner solutions where $\omega_0 = y'$.

For equal benefit to hold, removing the subsidy s —a pure income effect—must leave labor supply unchanged. To see this, set the endowment to $\omega = y'$ and select a' such that the agent chooses $(y', y' + s)$ under t_s :

$$y^u(y', a', t_s) = y' \quad \text{and} \quad x^u(y', a', t_s) = y' + s$$

(a corner solution with $l = 0$) (Figure 5).⁷ Now suppose that removing s generates an income effect that increases labor supply (equivalently, decreases leisure, since leisure is non-inferior). Thus,

$$y^u(y', a', 0) > y'.$$

⁷For every $\omega \geq 0$, there exists $a_\omega > 0$ such that

$$MRS_{(\omega, a_\omega)}^u(y', y' + s) = MRS_{(\omega_0, a)}^u(y', y' + s) \geq 1.$$

The utility gain from the subsidy at wage a' equals

$$u(l^u(y', a', t_s), x^u(y', a', t_s)) - u(l^u(y', a', 0), x^u(y', a', 0)).$$

Now decrease the wage rate from a' . Since the agent continues to choose $(y', y' + s)$ after the subsidy, post-subsidy utility remains unchanged while tax-free utility strictly falls—violating equal benefit.

We have shown that if t_s implements equal benefit among the poor, then (i) t_s has no kink point at $z - s$, and (ii) removing the subsidy s from any post-subsidy bundle $(y, y + s)$ with $y \leq z - s$ generates no income effect on labor supply. These conditions are satisfied by quasilinear preferences; identifying economically meaningful non-quasilinear examples that satisfy them poses an open question.

4.2.12. Poverty preservation

We introduce a fairness principle based on order consistency, analogous to the familiar rank-preservation requirement in standard tax design. Rank preservation within a society requires that taxation not reverse the income ordering of individuals: those who are richer pre-tax should not become poorer post-tax. Our requirement applies the same logic across distributions rather than individuals. If one society exhibits weakly less poverty than another before taxation, then taxation should not reverse that ordering afterward. Poverty preservation is thus the cross-society analogue of rank preservation: it rules out rank reversals in poverty standings across economies. Taxation may change poverty levels within each society, but it should not scramble pre-existing poverty hierarchies across societies.

In the endogenous income framework, this principle must be formalized with respect to the underlying fundamentals (ω, a, u) that generate income distributions. For a tax schedule t , let

$$x^u(\omega, a, t) = (x_1^{u_1}(\omega_1, a_1, t), \dots, x_n^{u_n}(\omega_n, a_n, t))$$

denote the post-tax income distribution induced by fundamentals (ω, a, u) .

Definition 2. A tax schedule t satisfies *poverty preservation* if for all vector fundamentals (ω, a, u) and (ω', a', u') with

$$x^{u'}(\omega', a', 0) \succ_z x^u(\omega, a, 0),$$

we have

$$x^{u'}(\omega', a', t) \succ_z x^u(\omega, a, t).$$

Poverty preservation implies poverty reduction.

Proposition 4. Any poverty-preserving tax schedule is $\succ_z$ -poverty reducing.

To see why **Proposition 4** holds, suppose that a tax schedule t is not $\succ_z$ -poverty reducing. By **Theorem 1**, such a schedule cannot be a lump-sum subsidy below the poverty line. Consequently, there exists some pre-tax income level below z at which the after-tax budget set, viewed in the (y, x) space of pre-tax and post-tax income, has slope strictly less than one. As illustrated in Step 3 of the informal proof of **Theorem 1** (see **Figure 3** in **Section 4.1**), this distortion implies that preferences can be found for which an individual optimally

chooses lower net consumption under taxation than in the tax-free benchmark. The proof of [Theorem 1](#) (see [Appendix A](#)) already establishes the existence of such preferences and primitives, so no further construction is required here.

Consider a one-person economy A exhibiting this behavior, so that consumption falls under t . Compare it with another one-person economy B , with different preferences, for which consumption remains above its tax-free level under t . The primitives can be selected so that initially A exhibits less poverty than B (higher tax-free consumption), while after taxation the consumption levels cross. The poverty ranking is thus reversed, contradicting poverty preservation, which requires pre-tax poverty orderings to be maintained after taxation.

Hence any tax schedule that fails to be $\succsim_z$ -poverty reducing cannot be poverty preserving.

The converse implication fails. Uniform lump-sum subsidies satisfy poverty reduction ([Theorem 1](#)) but may violate poverty preservation due to heterogeneous income effects. This tension is cleanly illustrated with two single-agent economies.

Consider two societies, each with one poor individual. In the tax-free environment, agent A consumes more than agent B , so society A exhibits strictly less poverty than society B . Now implement a uniform subsidy $s > 0$ that leaves both agents poor post-tax. Because preferences differ, agent A may reduce labor supply more aggressively than agent B in response to the income effect. If this behavioral response is sufficiently heterogeneous, post-tax income ranks reverse—reversing the poverty ranking.

Both agents still experience poverty reduction individually, satisfying the requirement of [Theorem 1](#). Yet the relative poverty positions of the two societies are scrambled: a policy that improves each society internally fails to preserve their pre-tax poverty ordering. This reversal occurs because heterogeneous behavioral responses distort the cross-society comparison.

Thus, $\succsim_z$ -poverty reduction does not entail poverty preservation. The former demands improvement *within* each society; the latter demands consistency *across* societies. When preferences vary, even the uniquely characterized poverty-reducing policies may satisfy the first requirement while violating the second—a tension that cannot be resolved without restricting the domain of admissible preferences.

5. Concluding remarks

This paper treats poverty reduction as an explicit non-utilitarian policy objective, a criterion that aligns with what [Kanbur et al. \(2006\)](#) identify as “behavioral public economics,” wherein the government optimizes based on objectives distinct from individuals’ utility functions. By shifting toward a concrete and widely recognized policy goal, this approach avoids the fundamental challenge of interpersonal utility comparisons in classical welfare economics. However, since poverty measures are typically income-based, they do not capture the individual disutility of work effort. From a positive perspective, this omission may be justified by the perceptions of taxpayers who finance poverty alleviation programs, as they may place greater emphasis on the income of the poor rather than their leisure choices ([Besley and Coate, 1995](#)).

This paper completely characterizes poverty-reducing nonlinear tax schedules within an endogenous income framework accommodating heterogeneous wage distributions, individual-specific non-labor endowments, and idiosyncratic preferences. When negative

marginal tax rates are deemed infeasible, poverty-reducing tax schedules take a simple form: a fixed lump-sum subsidy for after-tax incomes below the poverty line. These schedules admit a complete ordering by subsidy magnitude, with larger subsidies corresponding to greater poverty reduction. When negative marginal tax rates are permissible, poverty-reducing schedules must subsidize the poor at non-positive marginal rates.

Several extensions clarify the scope and boundaries of these results, revealing robustness to poverty line uncertainty as well as to endogenous poverty lines anchored to pre-determined reference statistics (e.g., lagged medians).

A comparison of our framework with its exogenous-income counterpart reveals how behavioral responses impose stringent structure through incentive compatibility. When income is fixed, virtually any subsidy to the poor suffices for poverty reduction. When income is endogenous, however, only policies with non-positive marginal tax rates on individuals below the poverty line guarantee poverty reduction—either eliminating substitution effects entirely (via uniform lump-sum subsidies) or generating pro-work substitution effects (via negative marginal rates).

The robustness of our characterization across arbitrary preference profiles stems directly from the Monotonicity axiom—embedded in censored first-order stochastic dominance—which treats income gains for the poor as unambiguously poverty-reducing. This Paretian principle enables a complete characterization using homogeneous economies alone, eliminating any need for preference restrictions. By contrast, distribution-sensitive poverty measures (which replace monotonicity with inequality concerns among the poor) cannot leverage homogeneous counterexamples, as relative deprivation depends on heterogeneous behavioral responses. Consequently, poverty reduction under such criteria requires restrictions on preference elasticities—a limitation absent from our framework.

As shown in [Section 4.2.5](#), while no tax schedule can universally reduce poverty under $\succ_R^z$ or $\succ_A^z$ across fully heterogeneous preferences, this impossibility vanishes under the commonly adopted assumption of homogeneous preferences. In that case, uniform lump-sum subsidies—or more generally, marginal-rate progressive schedules—can guarantee poverty reduction under certain wage elasticity conditions. This aligns with results in the literature on taxation and income inequality, suggesting that structural assumptions on preferences could yield implementable, distribution-sensitive anti-poverty policies.

Finally, an important open question is how our results extend to settings where poverty is defined multidimensionally—encompassing health, education, housing, or environmental quality alongside income. In such frameworks, a rise in income need not constitute poverty reduction if other dimensions of deprivation worsen simultaneously. It remains unclear whether a lump-sum subsidy, possibly conditioned on verifiable non-income indicators, can preserve both incentive compatibility and universal poverty reduction in this broader context. The interplay between multidimensional poverty orderings, behavioral responses across multiple margins, and the design of non-distortionary tax instruments presents a rich avenue for future research.

A. Proof of Theorem 1

We begin with a result on the pre-order $\succ_D$:

Lemma 1. *For any two income distributions $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{y} = (y_1, \dots, y_n)$, $x_i \geq y_i$ for each i implies $\mathbf{x} \succ_D \mathbf{y}$.*

Proof. By definition of $\succsim_D$, it suffices to show $x_{[k]} \geq y_{[k]}$ for all $k \in \{1, \dots, n\}$.

Suppose for contradiction that $x_{[k]} < y_{[k]}$ for some k . Define

$$A = \{i \in \{1, \dots, n\} : x_i \leq x_{[k]}\}.$$

By definition of the k -th order statistic, $|A| \geq k$. For each $i \in A$,

$$y_i \leq x_i \leq x_{[k]} < y_{[k]}.$$

Thus at least k elements of $\mathbf{y}$ are strictly less than $y_{[k]}$. But by definition of $y_{[k]}$ as the k -th smallest element, at most $k - 1$ elements of $\mathbf{y}$ can be strictly less than $y_{[k]}$ —a contradiction.

Hence $x_{[k]} \geq y_{[k]}$ for all k , so $\mathbf{x} \succsim_D \mathbf{y}$. $\blacksquare$

We establish as a preliminary result that all poverty-reducing tax schedules necessarily involve poor subsidization.

Lemma 2. Suppose that $t \in \mathcal{T}$ is $\succsim_z$ -poverty-reducing. Then $t(y) \leq 0$ for all $y \in [0, z]$.

Proof. Suppose that $t(x^*) > 0$ for some $x^* \in [0, z]$. We will show that t is not $\succsim_z$ -poverty-reducing.

Fix $a > x^*$ and define $\alpha = x^*/a$. Set the non-labor endowments equal to zero, $(\omega_1, \dots, \omega_n) = \mathbf{0}$, and pick the homogeneous wage rate distribution $(a, \dots, a)$ and the vector of identical preferences $(u, \dots, u)$, where

$$u(l, x) = x^\alpha (1 - l)^{1-\alpha}.$$

It is easy to see that the solution to the problem

$$\max_{y \in [0, a]} u(y/a, y),$$

which determines the individuals' gross income in the absence of taxation, is $x^u(0, a, 0) = x^*$.

When confronted with the tax schedule t , individuals choose their gross incomes to solve the problem

$$\max_{y \in [0, a]} u(y/a, y - t(y)). \quad (9)$$

If y^* denotes a solution to this problem and $y^* \leq x^*$, then the corresponding net income,

$$x^u(0, a, t) = y^* - t(y^*), \quad (10)$$

must satisfy $x^u(0, a, t) < x^*$. Indeed, this is clearly the case if $y^* = x^*$, since $t(x^*) > 0$ by assumption. If $y^* < x^*$, then

$$x^u(0, a, t) = y^* - t(y^*) \leq x^* - t(x^*) < x^*,$$

where the weak inequality holds because t preserves income ranks.

If $y^* > x^*$, the inequality $x^u(0, a, t) < x^*$ still holds. To see this, note first that, because y^* solves (9), there is a line $x = \beta y + \gamma$ with $\beta \leq 1$ that is tangent to the indifference curve

$$u(y/a, x) = u(y^*/a, y^* - t(y^*))$$

at $(y^*, y^* - t(y^*))$.

To see that the slope β is less than or equal to 1, consider first the case when y^* lies in the interior of a tax bracket. Then $x = \beta y + \gamma$ is the linear extension of the map from gross income to net income $y \mapsto y - t(y)$ restricted to a sufficiently small neighborhood of y^* . This restriction is linear with slope $1 - t'(y) \in [0, 1]$.

Alternatively, y^* may represent a kink point for the tax schedule t , where the marginal tax rate transitions from τ_0 to τ_1 . For this to be the case, we must have $\tau_0 < \tau_1$; otherwise, the solution y^* would not occur at the kink point. Moreover, the marginal rate of substitution at $(y^*, y^* - t(y^*))$,

$$MRS_a^u(y^*, y^* - t(y^*)) = -(1/a) \cdot \frac{\partial u(y^*/a, y^* - t(y^*))}{\partial l} \bigg/ \frac{\partial u(y^*/a, y^* - t(y^*))}{\partial x},$$

belongs to the interval $[1 - \tau_1, 1 - \tau_0]$. Since $\tau_0, \tau_1 \in [0, 1]$, the slope of the line $x = \beta y + \gamma$, β , which coincides with $MRS_a^u(y^*, y^* - t(y^*))$, is less than or equal to 1.

Now, since the line $x = \beta y + \gamma$ is tangent to the indifference curve

$$u(x, y/a) = u(y^* - t(y^*), y^*/a)$$

at $(y^*, y^* - t(y^*))$, y^* is also a solution to the problem

$$\max_{y \in [0, a]} u(y/a, \beta y + \gamma). \quad (11)$$

This implies that an individual with utility function u and wage rate a would choose the gross income-net income pair $(y^*, y^* - t(y^*))$ when faced with feasible consumption bundles of the form $(y, \gamma + \beta y)$.

Moreover, this chosen pair is less preferred than the pair (x^*, x^*) . This preference ordering can be established as follows:

- Due to the monotonicity of t , we have $0 < t(x^*) \leq t(y^*)$. Consequently, $(y^*, y^* - t(y^*))$ is less preferred than (y^*, y^*) .
- In turn, (y^*, y^*) is less preferred than (x^*, x^*) , since (x^*, x^*) is the optimal choice in the absence of taxation, while (y^*, y^*) remains available in that scenario.

This sequence of preferences demonstrates that the pair $(y^*, y^* - t(y^*))$ is indeed less preferred than (x^*, x^*) .

Consider now a scenario where the net income in problem (11) is augmented by a lump sum subsidy, e . This subsidy is precisely calibrated to enable the consumer to attain a bundle that yields the same utility as (x^*, x^*) . This lump sum addition represents a pure income effect. Let $(\hat{y}, \hat{x})$ denote an optimal gross income-net income bundle corresponding to the modified problem

$$\max_{y \in [0, a]} u(y/a, \beta y + \gamma + e),$$

where the feasible bundles now take the form $(y, \beta y + \gamma + e)$. Given our assumption that consumption is a normal good, we can assert that

$$\hat{x} > y^* - t(y^*), \quad (12)$$

Furthermore, we can establish that

$$\hat{x} \leq x^*. \quad (13)$$

This inequality can be demonstrated as follows:

- By construction, both $(\hat{y}, \hat{x})$ and (x^*, x^*) lie on the same indifference curve, indicating equal utility.
- The marginal rate of substitution at $(\hat{y}, \hat{x})$, $MRS_a^u(\hat{y}, \hat{x})$, is equal to β , which we previously established is less than or equal to 1.
- In contrast, $MRS_a^u(x^*, x^*) = 1$.
- Given that the marginal rate of substitution is increasing along an indifference curve, we can conclude that $\hat{y} \leq x^*$.
- Consequently, since both points lie on the same indifference curve and $\hat{y} \leq x^*$, it follows that $\hat{x} \leq x^*$.

Equations (10), (12), and (13) together yield

$$x^u(0, a, t) = y^* - t(y^*) < x^* = x^u(0, a, 0) \leq z,$$

implying that

$$(x^u(0, a, 0), \dots, x^u(0, a, 0)) \not\geq_z (x^u(0, a, t), \dots, x^u(0, a, t)).$$

Hence, t is not $\succsim_z$ -poverty-reducing. ■

Theorem 1. A tax schedule $t \in \mathcal{T}$ is $\succsim_z$ -poverty-reducing if and only if $t(y) = -s$ for all y with $y + s \leq z$ and some $s \geq 0$.

Proof. [$\Leftarrow$] Suppose that $t(y) = -s$ for all y with $y + s \leq z$ and some $s \geq 0$. We must show that t is $\succsim_z$ -poverty-reducing, meaning that for all endowment distributions $(\omega_1, \dots, \omega_n)$, all wage distributions $(a_1, \dots, a_n)$, all preference profiles $(u_1, \dots, u_n) \in \mathcal{U}^n$, and all solution functions $x_1^{u_1}, \dots, x_n^{u_n}$, we have

$$(x_1^{u_1}(\omega_1, a_1, 0), \dots, x_n^{u_n}(\omega_n, a_n, 0)) \succsim_z (x_1^{u_1}(\omega_1, a_1, t), \dots, x_n^{u_n}(\omega_n, a_n, t)). \quad (14)$$

For fixed parameters $(\omega_1, \dots, \omega_n)$, $(a_1, \dots, a_n)$, $(u_1, \dots, u_n)$, and solution functions $x_1^{u_1}, \dots, x_n^{u_n}$, let

$$c = (c_1, \dots, c_n) \quad \text{and} \quad c' = (c'_1, \dots, c'_n)$$

denote, respectively, the distributions

$$(x_1^{u_1}(\omega_1, a_1, 0), \dots, x_n^{u_n}(\omega_n, a_n, 0)) \quad \text{and} \quad (x_1^{u_1}(\omega_1, a_1, t), \dots, x_n^{u_n}(\omega_n, a_n, t))$$

censored at the poverty line z .

To see that (14) holds, it suffices to show that $c' \succsim_D c$, which is implied by the inequalities

$$c'_i \geq c_i, \quad i \in \{1, \dots, n\} \quad (15)$$

(Lemma 1).

To see that (15) holds, choose $i \in \{1, \dots, n\}$ and note that either $c'_i < z$ or $c'_i = z$. The second case is easy to handle, since

$$c'_i = z \geq c_i.$$

Suppose that $c'_i < z$. Then $c'_i = y'_i + s$, where y'_i is the solution to the problem

$$\max_{y_i \in [\omega_i, \omega_i + a_i]} u_i((y_i - \omega_i)/a_i, y_i + s). \quad (16)$$

In the absence of taxation, individual i 's income is the unique solution y_i^* to

$$\max_{y_i \in [\omega_i, \omega_i + a_i]} u_i((y_i - \omega_i)/a_i, y_i). \quad (17)$$

Letting

$$\tilde{y}_i = y_i - \omega_i,$$

(16) and (17) can be reformulated as

$$\max_{\tilde{y}_i \in [0, a_i]} u_i(\tilde{y}_i/a_i, \tilde{y}_i + \omega_i + s) \quad \text{and} \quad \max_{\tilde{y}_i \in [0, a_i]} u_i(\tilde{y}_i/a_i, \tilde{y}_i + \omega_i).$$

Note that the only difference between these two problems is the fixed subsidy s , which represents a pure income effect. Consequently, since consumption is a normal good by assumption, it follows that

$$z > c'_i \geq y_i^* = c_i,$$

as we sought.

[$\Rightarrow$] Suppose that t is $\succsim_z$ -poverty-reducing. We must show that $t(y) = -s$ for all y with $y + s \leq z$ and some $s \geq 0$.

Suppose, by contradiction, that t does not have this property. Then there exists a tax bracket overlapping with the set

$$\{y \in [0, z) : y - t(y) < z\}$$

where the marginal tax rate is strictly positive. If this were not true, the right derivative of t would be zero over the entire set $\{y \in [0, z) : y - t(y) < z\}$. This would imply that t is constant over this set and, by the continuity of t , constant over its closure $[0, \max\{y \in [0, z] : y - t(y) \leq z\}]$.

We analyze two exhaustive cases. In case (a), we consider tax brackets that overlap with the set of pre-tax incomes $\{y \in [0, z) : y - t(y) < z\}$, where at least one such bracket I has a marginal tax rate τ strictly between 0 and 1. In case (b), we also consider positive-rate brackets overlapping with this set, but we assume that all such brackets (of which there is at least one) have a marginal tax rate equal to 1.

First, we consider case (a). Select y' in the interior of I such that $y' < z$ and $y' - t(y') < z$. Furthermore, we can ensure that y' is chosen in a way that

$$y' - t(y') > y'. \quad (18)$$

This last condition is feasible because $t(y) \leq 0$ holds for all y in the interval $[0, z]$ (Lemma 2).

Consider the utility function

$$u(l, x) = \theta \left(\frac{x^{1-\alpha}}{1-\alpha} + \beta(1-l) \right) + (1-\theta) \left(\frac{(1-l)^{1-\alpha}}{1-\alpha} \right), \quad (19)$$

where $\alpha > 1$, $\beta > 0$, and $\theta \in (0, 1)$. This function belongs to the family of preferences $\mathcal{U}$.⁸

Since t is linear on the tax bracket I , and since $\tau \in (0, 1)$ is the marginal tax rate corresponding to I , we can write

$$t(y) = -b + \tau y, \quad \text{for all } y \in I,$$

for some $b \geq 0$.

An individual with a wage rate of $a > 0$ chooses, under the linear tax schedule $-b + \tau y$, an optimal gross income level that solves the following optimization problem:

$$\max_{y \in [0, a]} u(y/a, b + (1-\tau)y).$$

Because the objective function is strictly concave, an interior solution to this problem exists if and only if

$$\theta(1-\tau)(b + (1-\tau)y)^{-\alpha} = \frac{\theta\beta}{a} + \frac{1-\theta}{a}(1-y/a)^{-\alpha}.$$

Fixing $a > y' > 0$, plugging $y = y' \in (0, a)$ into this equation, and solving for β yields

$$\beta = a(1-\tau)(b + (1-\tau)y')^{-\alpha} - \frac{1-\theta}{\theta}(1-y'/a)^{-\alpha}. \quad (20)$$

Note that for any $\alpha > 1$, and for any θ sufficiently close to 1, the corresponding β determined by (20) satisfies $\beta > 0$, ensuring it lies within the feasible parameter range.

Consequently, for any $\alpha > 1$, there exists $\theta_\alpha \in (0, 1)$ such that for every $\theta \in (\theta_\alpha, 1)$ we have $\beta(\alpha, \theta) > 0$, where

$$\beta(\alpha, \theta) = a(1-\tau)(b + (1-\tau)y')^{-\alpha} - \frac{1-\theta}{\theta}(1-y'/a)^{-\alpha},$$

⁸The utility function u is continuous over its domain $\mathbb{R}_+ \times [0, 1]$. Consider the marginal rate of substitution of consumption for leisure in the interior of this domain:

$$MRS^u(l, x) = -\frac{\partial u(l, x)}{\partial l} \bigg/ \frac{\partial u(l, x)}{\partial x} = x^\alpha \left(\beta + \frac{1-\theta}{\theta}(1-l)^{-\alpha} \right)$$

This MRS is strictly increasing along any indifference curve, which establishes that u is strictly quasiconcave in the interior of $\mathbb{R}_+ \times [0, 1]$. Furthermore, we can verify that consumption is a normal good by examining the partial derivatives of the MRS:

$$\frac{\partial MRS^u(l, x)}{\partial l} = x^\alpha \cdot \frac{\alpha(1-\theta)}{\theta}(1-l)^{-\alpha-1} > 0 \quad \text{and} \quad \frac{\partial MRS^u(l, x)}{\partial x} = \alpha x^{\alpha-1} \left[\beta + \frac{1-\theta}{\theta}(1-l)^{-\alpha} \right] > 0.$$

Both partial derivatives are strictly positive under our parameter assumptions, confirming that consumption is indeed a normal good.

and the optimization problem

$$\max_{y \in [0, a]} \theta \left(\frac{(b + (1 - \tau)y)^{1-\alpha}}{1 - \alpha} + \beta(\alpha, \theta)(1 - y/a) \right) + (1 - \theta) \left(\frac{(1 - y/a)^{1-\alpha}}{1 - \alpha} \right) \quad (21)$$

admits y' as its unique solution.

Next, consider the behavior of the indifference curve passing through the gross income-net income bundle $(y', b + (1 - \tau)y')$. The utility equality along this curve is given by

$$\begin{aligned} \theta \left(\frac{x^{1-\alpha}}{1 - \alpha} + \beta(\alpha, \theta) \left(1 - \frac{y}{a} \right) \right) + (1 - \theta) \left(\frac{(1 - y/a)^{1-\alpha}}{1 - \alpha} \right) \\ = \theta \left(\frac{(b + (1 - \tau)y')^{1-\alpha}}{1 - \alpha} + \beta(\alpha, \theta) \left(1 - \frac{y'}{a} \right) \right) + (1 - \theta) \left(\frac{(1 - y'/a)^{1-\alpha}}{1 - \alpha} \right). \end{aligned}$$

The marginal rate of substitution at any point (y, x) on this indifference curve is:

$$\begin{aligned} \left(-\frac{\partial u(y/a, x)}{\partial y} \middle/ \frac{\partial u(y/a, x)}{\partial x} \right) \bigg|_{\beta=\beta(\alpha, \theta)} &= (1 - \tau) \left(\frac{x}{b + (1 - \tau)y'} \right)^\alpha \\ &\quad + \frac{x^\alpha}{a} \frac{1 - \theta}{\theta} \left[(1 - y/a)^{-\alpha} - (1 - y'/a)^{-\alpha} \right], \end{aligned}$$

where this expression is positive when $\beta(\alpha, \theta) > 0$.

For sufficiently large α and any correspondingly large $\theta \in (\theta_\alpha, 1)$, the marginal rate of substitution at any bundle (y, x) on the indifference curve can be made arbitrarily close to zero for $x < b + (1 - \tau)y'$ and arbitrarily large for $x > b + (1 - \tau)y'$.

This property ensures that for sufficiently large α and $\theta \in (\theta_\alpha, 1)$, the gross income-net income bundle

$$(y', y' - t(y')) = b + (1 - \tau)y'$$

is also optimal under the tax schedule t . The indifference curve through this bundle becomes arbitrarily steep for gross income levels to the right of y' and arbitrarily flat for levels to the left of y' , making all other feasible bundles under tax schedule t less preferred.

Therefore, the solution to (21) coincides with the solution to the same problem where the linear tax schedule $b + \tau y$ is replaced by t :

$$\max_{y \in [0, a]} \theta \left(\frac{(y - t(y))^{1-\alpha}}{1 - \alpha} + \beta(\alpha, \theta) \left(1 - \frac{y}{a} \right) \right) + (1 - \theta) \left(\frac{(1 - y/a)^{1-\alpha}}{1 - \alpha} \right).$$

Next, observe that because

$$\theta(1 - \tau)(y' - t(y'))^{-\alpha} = \frac{\theta\beta(\alpha, \theta)}{a} + \frac{1 - \theta}{a}(1 - y'/a)^{-\alpha},$$

and since $1 - \tau < 1$, for sufficiently large $\theta \in (0, 1)$ we have

$$\theta(y' - t(y'))^{-\alpha} > \frac{\theta\beta(\alpha, \theta)}{a} + \frac{1 - \theta}{a} \left(1 - \frac{y' - t(y')}{a} \right)^{-\alpha}.$$

The expression $\theta y^{-\alpha}$ is decreasing in y while

$$\frac{\theta\beta(\alpha, \theta)}{a} + \frac{1 - \theta}{a} \left(1 - \frac{y}{a}\right)^{-\alpha}$$

is increasing in y . Moreover, for y sufficiently close to a , we have

$$\theta y^{-\alpha} < \frac{\theta\beta(\alpha, \theta)}{a} + \frac{1 - \theta}{a} \left(1 - \frac{y}{a}\right)^{-\alpha}.$$

By the Intermediate Value Theorem, there exists $y^* \in (y' - t(y'), a)$ satisfying

$$\theta y^{*- \alpha} = \frac{\theta\beta(\alpha, \theta)}{a} + \frac{1 - \theta}{a} \left(1 - \frac{y^*}{a}\right)^{-\alpha}.$$

This y^* is precisely the unique solution to the first-order condition of the tax-free optimization problem:

$$\max_{y \in [0, a]} \theta \left(\frac{y^{1-\alpha}}{1-\alpha} + \beta(\alpha, \theta)(1 - y/a) \right) + (1 - \theta) \left(\frac{(1 - y/a)^{1-\alpha}}{1-\alpha} \right).$$

Therefore, for sufficiently large α and θ ,

$$x^u(0, a, 0) = y^* > y' - t(y') = x^u(0, a, t), \quad (22)$$

where the second inequality follows from (18).

The crucial step in reaching our contradiction is as follows: we have constructed a utility function $u \in \mathcal{U}$ for which (22) holds. This implies that an agent with utility function u and wage rate a achieves lower after-tax income under tax regime t compared to a tax-free environment. Moreover, this lower after-tax income, y' , satisfies

$$y' < y' - t(y') < z.$$

Consequently,

$$(x^u(0, a, 0), \dots, x^u(0, a, 0)) \not\preceq_z (x^u(0, a, t), \dots, x^u(0, a, t)).$$

This directly contradicts our assumption that t is $\succsim_z$ -poverty-reducing, completing the proof.

It remains to consider case (b). Recall that in this case t has at least one tax bracket overlapping with the set $\{y \in [0, z] : y - t(y) < z\}$ where the marginal tax rate is positive. Moreover, all such overlapping brackets have a marginal tax rate of 1.

To begin, we show that the intercept $t(0)$ must be negative. If instead $t(0) \geq 0$, then since $t(y) \leq 0$ for all $y \in [0, z]$ (Lemma 2), the monotonicity of t would imply $t(y) = 0$ for all $y \in [0, z]$. This would mean t has the property that $t(y) = -s$ for some $s \geq 0$ whenever $y + s \leq z$, contradicting our assumption.

Given $t(0) < 0$, there are two possible subcases: (i) the first tax bracket has a marginal tax rate of 1; and (ii) the first tax bracket has a zero marginal tax rate, and at some point $\hat{y} < z$ where $\hat{y} - t(\hat{y}) < z$, a second tax bracket begins with a marginal tax rate of 1.

The proof technique from case (a) can be adapted to handle both subcases, with appropriate modifications. Since the arguments are similar, we present only subcase (i) in detail.

Let $-\gamma = t(0)$ to simplify notation. Note that $\gamma < z$, otherwise, by rank preservation of t , there would be no overlap of at least one tax bracket with the set $\{y \in [0, z) : y - t(y) < z\}$.

We return to the utility function specified in (19) and analyze the optimization problem faced by an agent with wage rate a under a linear tax schedule $-\gamma + \lambda y$, where $\lambda \in (0, 1)$:

$$\max_{y \in [0, a]} \theta \left(\frac{(\gamma + (1 - \lambda)y)^{1-\alpha}}{1 - \alpha} + \beta(1 - y/a) \right) + (1 - \theta) \left(\frac{(1 - y/a)^{1-\alpha}}{1 - \alpha} \right). \quad (23)$$

Observe that $y = 0$ emerges as the unique solution to (23) whenever

$$\theta(1 - \lambda)\gamma^{-\alpha} \leq \frac{\theta\beta}{a} + \frac{1 - \theta}{a}.$$

Consequently, for fixed $a > 0$ and any $\alpha > 1$, there exists $\bar{\theta} \in (0, 1)$ such that for every $\theta \in (\bar{\theta}, 1)$, we have $\beta(\alpha, \theta) > 0$, where

$$\beta(\alpha, \theta) = a(1 - \lambda)\gamma^{-\alpha} - \frac{1 - \theta}{\theta}, \quad (24)$$

and the optimization problem

$$\max_{y \in [0, a]} \theta \left(\frac{(\gamma + (1 - \lambda)y)^{1-\alpha}}{1 - \alpha} + \beta(\alpha, \theta)(1 - y/a) \right) + (1 - \theta) \left(\frac{(1 - y/a)^{1-\alpha}}{1 - \alpha} \right) \quad (25)$$

admits $y = 0$ as its unique solution.

Following the reasoning from case (a), we can establish via revealed preference that for sufficiently large α and θ , the solution to (25) coincides with the solution to the analogous problem where the linear tax schedule $-\gamma + \lambda y$ is replaced by t :

$$\max_{y \in [0, a]} \theta \left(\frac{(y - t(y))^{1-\alpha}}{1 - \alpha} + \beta(\alpha, \theta)(1 - y/a) \right) + (1 - \theta) \left(\frac{(1 - y/a)^{1-\alpha}}{1 - \alpha} \right). \quad (26)$$

This equivalence follows from two observations. First, the constraint set of gross income-net income pairs for problem (26) is strictly contained within that of problem (25) in a neighborhood of 0. Second, for sufficiently large α , this neighborhood of 0 becomes decisive for the agent's optimization in both problems, as all feasible gross income levels outside this region become strictly inferior choices. Since the pair $(y = 0, x = \gamma)$ is attainable in both problems and (25) permits a wider range of choices near $y = 0$, we can conclude that $y = 0$ must be optimal for both problems.

Note that (24) implies that for sufficiently large α and θ ,

$$\theta(1 - \lambda)\gamma^{-\alpha} = \frac{\theta\beta(\alpha, \theta)}{a} + \frac{1 - \theta}{a}.$$

Moreover, since $1 - \lambda < 1$, for sufficiently large $\theta \in (0, 1)$ we have

$$\theta \gamma^{-\alpha} > \frac{\theta \beta(\alpha, \theta)}{a} + \frac{1 - \theta}{a} \left(1 - \frac{\gamma}{a}\right)^{-\alpha}.$$

Applying the Intermediate Value Theorem as in the previous case yields the existence of $y^* \in (\gamma, a)$ satisfying

$$\theta y^{*-\alpha} = \frac{\theta \beta(\alpha, \theta)}{a} + \frac{1 - \theta}{a} \left(1 - \frac{y^*}{a}\right)^{-\alpha}.$$

This y^* uniquely solves the first-order condition of the tax-free optimization problem:

$$\max_{y \in [0, a]} \theta \left(\frac{y^{1-\alpha}}{1-\alpha} + \beta(\alpha, \theta)(1 - y/a) \right) + (1 - \theta) \left(\frac{(1 - y/a)^{1-\alpha}}{1-\alpha} \right).$$

Thus, for sufficiently large α and θ ,

$$x^u(0, a, 0) = y^* > \gamma = x^u(0, a, t),$$

which leads to the same contradiction as in case (a), since $\gamma < z$. ■

B. Relaxing monotonicity

This section provides a complete characterization of poverty-reducing tax schedules in the absence of the monotonicity constraint, extending the results of [Theorem 1](#) to this more general setting.

For the remainder of this section, a *tax schedule* is defined as a continuous and rank-preserving function $t : \mathbb{R}_+ \rightarrow \mathbb{R}$, where $t(y)$ represents the tax liability corresponding to the pre-tax income level y .

Note that dropping the monotonicity assumption renders tax schedules flexible enough to accommodate negative marginal tax rates.

Let $\mathcal{T}^*$ denote the set of all tax schedules according to this weaker definition.

We first state and prove the analogue of [Lemma 2](#).

Lemma 3. *Suppose that $t \in \mathcal{T}^*$ is $\succsim_z$ -poverty-reducing. Then $t(y) \leq 0$ for all $y \in [0, z]$.*

Proof. Suppose there exists some $x^* \in [0, z]$ where $t(x^*) > 0$. We will prove that t cannot be $\succsim_z$ -poverty-reducing.

To establish this, it suffices to demonstrate the existence of a utility function $u \in \mathcal{U}$ and a positive constant $a > 0$ such that

$$z > x^u(0, a, t) < x^u(0, a, 0). \tag{27}$$

Indeed, this inequality immediately implies

$$(x^u(0, a, 0), \dots, x^u(0, a, 0)) \not\succsim_z (x^u(0, a, t), \dots, x^u(0, a, t)),$$

thereby proving that t is not $\succsim_z$ -poverty-reducing.

First, note that there is no loss of generality in assuming that x^* belongs to the interior of a tax bracket.

Fix $a > x^*$ and define the constant elasticity of substitution (CES) utility function

$$u(l, x) = \left((1-l)^\rho + \beta^\rho x^\rho \right)^{\frac{1}{\rho}},$$

where $\rho < 0$ and $\beta > 0$.

If an individual with a wage rate of a were to choose among all the gross income-net income pairs of the form

$$(y, \eta + (1 + \gamma)y),$$

where $\eta = -t(x^*) - \gamma x^*$ and $\gamma > 0$, the optimal gross income level would solve the following optimization problem:

$$\max_{y \in [0, a]} u(1 - y/a, \eta + (1 + \gamma)y). \quad (28)$$

Here, the parameter η has been chosen so that the line $\eta + (1 + \gamma)y$ passes through the point $(x^*, x^* - t(x^*))$.

Note that γ can be chosen large enough to ensure that

$$z \geq y > x^* \Rightarrow \eta + (1 + \gamma)y > y - t(y). \quad (29)$$

Now consider the behavior of the utility function u as the parameter ρ converges to $-\infty$. The limit function is the Leontief-type utility function $\min\{1 - l, \beta x\}$, whose L -shaped indifference curves have kink points lying on the ray

$$x = \frac{1}{\beta}(1 - y/a).$$

This ray passes through the point $(x^*, x^* - t(x^*))$ provided that

$$\beta = (1 - x^*/a)/(x^* - t(x^*)). \quad (30)$$

For this β , the ray intersects with the 45° line at

$$\left(\frac{a(x^* - t(x^*))}{a - t(x^*)}, \frac{a(x^* - t(x^*))}{a - t(x^*)} \right).$$

Under (30), the solution to the tax-free optimization problem

$$\max_{y \in [0, a]} u(1 - y/a, y)$$

converges to $y = \frac{a(x^* - t(x^*))}{a - t(x^*)}$ as $\rho \rightarrow -\infty$, whereas the solution to (28) converges to $y = x^*$ as $\rho \rightarrow -\infty$.

Moreover, by the condition (29), no solution to the problem

$$\max_{y \in [0, a]} u(1 - y/a, y - t(y))$$

can possibly exceed the solution to (28).

Consequently, given (29) and (30), and for any sufficiently negative ρ , there exists a small enough $\epsilon > 0$ such that

$$x^u(0, a, t) \leq x^* - t(x^*) + \epsilon < \min \left\{ z, \frac{a(x^* - t(x^*))}{a - t(x^*)} \approx x^u(0, a, 0) \right\},$$

which yields the desired contradiction, (27). $\blacksquare$

Theorem 2. A tax schedule $t \in \mathcal{T}^*$ is $\succsim_z$ -poverty-reducing if and only if $t(y) \leq 0$ for all $y \in [0, z]$ and t is non-increasing over the set $[0, \max\{y \in [0, z] : y - t(y) \leq z\}]$.

Remark 4. Note that if a tax schedule is non-increasing over an interval $[0, \bar{y}]$ for some $\bar{y} > 0$, then t is rank-preserving over $[0, \bar{y}]$.

Proof of Theorem 2. [$\Leftarrow$] Suppose that $t(y) \leq 0$ for all $y \in [0, z]$ and t is non-increasing over the set $[0, \max\{y \in [0, z] : y - t(y) \leq z\}]$. We must show that t is $\succsim_z$ -poverty-reducing, meaning that for all endowment distributions $(\omega_1, \dots, \omega_n)$, all wage distributions $(a_1, \dots, a_n)$, all preference profiles $(u_1, \dots, u_n) \in \mathcal{U}^n$, and all solution functions $x_1^{u_1}, \dots, x_n^{u_n}$, we have

$$(x_1^{u_1}(\omega_1, a_1, 0), \dots, x_n^{u_n}(\omega_n, a_n, 0)) \succsim_z (x_1^{u_1}(\omega_1, a_1, t), \dots, x_n^{u_n}(\omega_n, a_n, t)). \quad (31)$$

For fixed parameters $(\omega_1, \dots, \omega_n)$, $(a_1, \dots, a_n)$, $(u_1, \dots, u_n)$, and solution functions $x_1^{u_1}, \dots, x_n^{u_n}$, let

$$c = (c_1, \dots, c_n) \quad \text{and} \quad c' = (c'_1, \dots, c'_n)$$

denote, respectively, the distributions

$$(x_1^{u_1}(\omega_1, a_1, 0), \dots, x_n^{u_n}(\omega_n, a_n, 0)) \quad \text{and} \quad (x_1^{u_1}(\omega_1, a_1, t), \dots, x_n^{u_n}(\omega_n, a_n, t))$$

censored at the poverty line z .

To see that (31) holds, it suffices to show that $c' \succsim_D c$, which is implied by the inequalities

$$c'_i \geq c_i, \quad i \in \{1, \dots, n\} \quad (32)$$

(Lemma 1).

To see that (32) holds, choose $i \in \{1, \dots, n\}$ and note that either $c'_i < z$ or $c'_i = z$.

The second case is easy to handle, since

$$c'_i = z \geq c_i.$$

Suppose that $c'_i < z$. Then $c'_i = y'_i - t(y'_i)$, where y'_i solves

$$\max_{y_i \in [\omega_i, \omega_i + a_i]} u_i((y_i - \omega_i)/a_i, y_i - t(y_i)). \quad (33)$$

In the absence of taxation, individual i 's income is the unique solution y_i^* to

$$\max_{y_i \in [\omega_i, \omega_i + a_i]} u_i((y_i - \omega_i)/a_i, y_i). \quad (34)$$

It suffices to show that

$$y'_i - t(y'_i) \geq y_i^*. \quad (35)$$

First, observe that $y'_i \leq z$ must hold since $y'_i - t(y'_i) < z$ and $t(y) \leq 0$ for each $y \in [0, z]$ (Lemma 3); otherwise, t would violate the rank-preservation property.

Given that $y'_i \leq z$ and $t(y) \leq 0$ for $y \in [0, z]$, individual i achieves a higher indifference curve when solving problem (33) with solution y'_i compared to the solution y_i^* of problem (34).

We can therefore introduce a lump sum subsidy $b > 0$ to problem (34) that is precisely large enough for individual i to reach the same indifference curve achieved at y'_i . This pure income effect transforms the optimization problem to:

$$\max_{y_i \in [\omega_i, \omega_i + a_i]} u_i((y_i - \omega_i)/a_i, y_i + b). \quad (36)$$

Let $(\hat{y}_i, \hat{y}_i + b)$ denote a solution to (36). Since consumption is a normal good, we have

$$y_i^* \leq \hat{y}_i + b. \quad (37)$$

Moreover, since t is non-increasing over $[0, \max\{y \in [0, z] : y - t(y) \leq z\}]$ and $y'_i - t(y'_i) < z$, the budget line slope in problem (36) is less than or equal to that in problem (33). As both $(\hat{y}_i, \hat{y}_i + b)$ and $(y'_i, y'_i - t(y'_i))$ lie on the same indifference curve, we can compare their marginal rates of substitution:

$$\begin{aligned} MRS_{(\omega_i, a)}^{u_i}(y'_i, y'_i - t(y'_i)) &= -(1/a) \cdot \frac{\partial u((y'_i - \omega_i)/a, y'_i - t(y'_i))}{\partial l} \bigg/ \frac{\partial u((y'_i - \omega_i)/a, y'_i - t(y'_i))}{\partial x} \\ &\geq -(1/a) \cdot \frac{\partial u((\hat{y}_i - \omega_i)/a, \hat{y}_i + b)}{\partial l} \bigg/ \frac{\partial u((\hat{y}_i - \omega_i)/a, \hat{y}_i + b)}{\partial x} = MRS_{(\omega_i, a)}^{u_i}(\hat{y}_i, \hat{y}_i + b). \end{aligned}$$

This implies $y'_i - t(y'_i) \geq \hat{y}_i + b$, which together with (37) establishes (35), completing our proof.

[$\Rightarrow$] Suppose that t is $\succ_z$ -poverty-reducing. By Lemma 3, $t(y) \leq 0$ for all $y \in [0, z]$, so we only need to show that t is non-increasing over the set $[0, \max\{y \in [0, z] : y - t(y) \leq z\}]$.

Suppose, by contradiction, that t does not have this property. Then there exists a tax bracket overlapping with the set

$$\{y \in [0, z] : y - t(y) < z\}$$

where the marginal tax rate is strictly positive. If this were not true, the right derivative of t would be zero over the entire set $\{y \in [0, z] : y - t(y) < z\}$. This would imply that t is constant over this set and, by the continuity of t , constant over its closure $[0, \max\{y \in [0, z] : y - t(y) \leq z\}]$.

As in the proof of Theorem 1, we consider two main cases. Case (a) examines tax brackets intersecting with the set $\{y \in [0, z] : y - t(y) < z\}$, where at least one bracket has a marginal tax rate strictly between 0 and 1. The argument deriving a contradiction for this case proceeds exactly as in the proof of Theorem 1.

Case (b) considers positive-rate brackets intersecting with the same set, but assumes all such brackets (of which at least one exists) have a marginal tax rate of 1. This case is further divided into two subcases: (i) the first tax bracket has a marginal tax rate of 1; and (ii) the first k tax brackets have a non-positive marginal tax rate, and at some point $\hat{y} < z$ where $\hat{y} - t(\hat{y}) < z$, the k -th tax bracket begins with a marginal tax rate of 1.

In subcase (i), because $t(y) \leq 0$ for all $y \in [0, z]$ (Lemma 3), we must have $t(0) < 0$, as in the proof of Theorem 1. This case can be resolved using the same argument as in that proof.

Subcase (ii) differs from Theorem 1, where case (b) necessarily implied $t(0) < 0$ due to the exclusion of negative marginal tax rates. The current framework, allowing such rates, cannot rule out the equality $t(0) = 0$. However, this subcase remains qualitatively similar to subcase (i) and can be handled with an analogous argument, which we omit for brevity. ■

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